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Accuracy, precision, recall, f1-score, or MCC? empirical evidence from advanced statistics, ML, and XAI for evaluating business predictive models

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Abstract

Imbalanced datasets pose a persistent challenge in business data mining, particularly in high-stakes domains such as financial risk prediction and customer churn analysis, where the minority class often carries disproportionate operational and financial consequences. Although widely used evaluation metrics—such as accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC)—are commonly applied in practice, there remains no empirical consensus on which metric offers the most reliable performance under real-world conditions. Existing studies lack a unified, statistically validated framework that accounts for threshold sensitivity, input noise, and interpretability—factors critical to business decision-making. To address this gap, we present a comprehensive and statistically rigorous evaluation of performance metrics for imbalanced business classification tasks. Using two benchmark datasets with distinct sizes and imbalance ratios—the Default of Credit Card Clients dataset and the Telco Customer Churn dataset—we evaluate five commonly used machine learning models: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and k-Nearest Neighbors (KNN). Our methodology incorporates static and dynamic threshold analysis, Gaussian noise robustness testing, bootstrap confidence intervals, McNemar's test, Cohen's kappa, and analysis of variance (ANOVA) to assess the statistical reliability of performance metrics. In addition, we introduce a novel two-stage explainable artificial intelligence (XAI) framework using SHapley Additive exPlanations (SHAP). The first stage employs standard SHAP visualizations (bar and beeswarm plots) to ensure baseline interpretability. The second stage extends this with a novel 3D metric-conditioned SHAP analysis, linking feature contributions to variations in classification thresholds and evaluation metrics. Our findings show that the F1-score consistently provides the most stable and balanced evaluation across datasets and testing conditions, with MCC offering complementary diagnostic value. In contrast, accuracy and precision demonstrate limited robustness under class imbalance. By combining statistical rigor with interpretable AI, this study offers the most comprehensive guidance to date for

selecting performance metrics in imbalanced business classification, with practical implications for model deployment in finance, marketing, and customer analytics.

Keywords Business, Data mining, Machine learning, Imbalanced datasets, Explainable AI (XAI), Statistics, Accuracy, Precision, F1, MCC

Introduction

In contemporary business environments, leveraging machine learning (ML) for decision-making has become indispensable. Applications in customer churn prediction, fraud detection, and financial credit scoring often rely on predictive models trained on real-world datasets [1]. However, such datasets are frequently imbalanced, where instances of one class significantly outnumber the other. This imbalance poses substantial challenges to model performance and evaluation, as standard accuracy-focused metrics fail to adequately assess predictive power across classes. Metrics like precision, recall, and MCC are better suited to these scenarios but are underutilized in business data mining applications [2]. Consequently, addressing imbalances effectively while optimizing performance metrics is critical for deploying reliable ML models in business settings. Existing studies have advanced our understanding of metric applicability but have significant limitations. For instance, [3] demonstrated the inadequacies of accuracy in imbalanced datasets but did not explore alternative metrics in depth. Similarly, [4] advocated the use of MCC for binary classification but lacked a comparative evaluation of MCC against other metrics like F1-score and precision. [5] optimized oversampling techniques for imbalanced data but neglected to consider dynamic thresholds or noise robustness. Moreover, [6] demonstrated the potential of oversampling for yield estimation in industrial applications but did not extend their methodology to broader business datasets. Finally, [7] highlighted the effectiveness of Area Under the Precision-Recall Curve (AUPRC) optimization for imbalanced datasets but did not integrate explainable AI (XAI) tools to interpret model outputs. Despite these advances in the current literature, several gaps remain. First, a comprehensive evaluation of many performance indicators encompassing accuracy, precision, recall, F1, and MCC is still unexplored in the business domain. Second, the implementation of advanced statistical techniques, such as ANOVA and McNemar's tests, for optimal metric selection is still underexplored. Second, the implementation of dynamic and static threshold sensitivity analysis has not been systematically examined in business data mining. Third, while noise robustness testing has gained traction in other domains, its application to metric evaluation in business data mining remains completely unexplored. Lastly, the implementation of XAI techniques like SHAP for metric-based feature impact analysis is still unexplored in current literature of business data mining. To address the identified gaps in the current literature on business data mining under class imbalance, the key contributions of our study are as follows:

- We conduct a comprehensive statistical validation of five most commonly used performance metrics using ANOVA, McNemar's test, and bootstrap confidence intervals to quantify both inter-metric and inter-model differences under imbalanced conditions.

- We implement both static and dynamic threshold sensitivity analyses to examine how classification thresholds affect each metric, providing a nuanced understanding of threshold dependence.
- We evaluate the robustness of performance metrics under varying levels of synthetic noise to simulate real-world data imperfections, identifying metrics most resilient to instability.
- We introduce a novel two-stage explainable AI (XAI) framework using SHAP. The first stage applies conventional SHAP plots for global and local interpretability, while the second leverages 3D metric-conditioned SHAP visualizations to reveal how feature importance interacts with threshold variation and metric outcomes.

To shed light on the above-mentioned contributions, our investigation is guided by the following research questions (RQs):

- **RQ1:** Which performance metrics among accuracy, precision, recall, F1-score, and MCC provide the most reliable evaluation for imbalanced classification in business domains?
- **RQ2:** How do static and dynamic threshold sensitivity analyses impact metric outcomes, and how can they be used to determine optimal decision boundaries?
- **RQ3:** How can advanced statistical methods such as ANOVA, McNemar's test, and bootstrap confidence intervals be used to validate and compare metric robustness?
- **RQ4:** How does input noise affect the stability of different metrics, and which metric remains most resilient under noisy, real-world business conditions?
- **RQ5:** How can explainable AI techniques like SHAP help interpret metric behavior across features and thresholds, and what actionable insights do they provide for model evaluation?

The reminder sections of this manuscript are structured as follows: Section Literature Review reviews related work, highlighting gaps in the literature. Section Methodology details the methodology, including dataset preprocessing, modeling, and analytical techniques. Section Experimental Results presents the experimental results, integrating statistical insights with XAI findings. Section Discussion presents the discussion and limitations. Finally, Section Conclusion outlines conclusions and proposes future research directions. With the problem clearly defined in the introduction, we now review the related literature to identify prevailing methodologies and highlight unresolved questions.

Literature review

Imbalanced datasets pose a persistent challenge in business data mining, especially in high-stakes domains such as credit risk, churn prediction, and fraud detection. In these settings, minority classes often represent critical outcomes such as defaults, cancellations, and anomalies, yet their underrepresentation leads to biased models and distorted performance evaluations [8]. Traditional metrics like accuracy frequently fail under such conditions, as highlighted by [9] and empirically supported by [10], both of whom recommend alternatives like F1-score and MCC for more balanced evaluation. To mitigate sampling bias, [11] proposed one-sided selection, while [12] confirmed MCC and F1-score as the most stable across varying imbalance ratios in health data.

Despite this, many domain-specific studies still rely on suboptimal practices. For example, [13] evaluated fraud detection using AUC and precision alone, without statistical or explainability validation, typifying a broader trend where metric selection is assumed rather than empirically supported. Similarly, studies like [14–17] rely solely on accuracy, while others such as [18–20] focus only on precision or recall, omitting metrics like MCC and F1 and lacking any statistical or explainability analysis.

Even among works using more robust metrics—such as [21] (F1, AUC), [22–24] (MCC)—comparative analysis and validation remain absent. Recent applications in churn prediction and financial modeling [25–28] likewise neglect optimal metric selection, threshold sensitivity, and interpretability via XAI. [29] evaluated metrics such as MCC and G-Mean in software defect prediction tasks, finding them to be more robust than accuracy or AUC. However, the study did not include threshold-based evaluation or any explainability component. A simulation-based comparison by [30] further analyzed twelve metrics, including MCC and Cohen’s Kappa, highlighting the superior reliability of MCC under imbalance but lacking any XAI integration. Additional works such as [27, 28] use multiple metrics (MCC, AUC) but omit statistical or interpretability validation. Foundational works like [31, 32] discuss ROC limitations and metric bias but do not explore metric interactions or XAI.

In contrast, explainable AI (XAI) has increasingly gained attention for enhancing model transparency in high-stakes domains. For example, [33] integrated SHAP with embedded feature selection in electricity demand forecasting, demonstrating its value in improving interpretability. [34] proposed a novel metric to validate SHAP-based explanations in time series forecasting, while [35] showcased SHAP’s applicability in healthcare for sleep apnea prediction. Additionally, [36] provided a comprehensive taxonomy of XAI evaluation techniques, though they noted the lack of standardized metrics and real-world validations. Despite these advancements, existing studies often treat SHAP explanations and metric evaluation as separate concerns.

Unlike the study by [37], who examined the consistency of four metrics (AUC, F-measure, G-mean, MCC) under varying imbalance ratios using correlation and simulation, our study extends far beyond theoretical consistency analysis. We evaluate five metrics across real-world business datasets using rigorous statistical tests (ANOVA, McNemar’s test, bootstrap CI), incorporate static and dynamic threshold sensitivity analysis, and assess metric robustness under input noise. Whereas their analysis considered metrics in isolation, our framework explores how metric behavior evolves across model types, input noise, threshold dynamics, and explainability layers. Most notably, we introduce a novel two-stage XAI framework, culminating in 3D metric-conditioned SHAP visualizations that link feature contributions to metric behavior across thresholds. Our work provides a unified, statistically grounded, threshold-aware, and explainability-enhanced approach to performance evaluation, offering greater practical utility than existing consistency-focused studies.

The limitations of existing business studies in handling metric evaluation under class imbalance are summarized in Table 1, while Table 2 presents recent studies focusing on evaluation metrics and explainability under imbalanced learning. Collectively, these studies underscore the absence of a unified, statistically validated, and explainability-aware framework for evaluating performance metrics under class imbalance. To bridge this gap, we evaluate five machine learning models on two real-world imbalanced

Table 1 Empirical business studies using evaluation metrics under class imbalance and identified gaps

Ref.	Business dataset used	Metric used	Model used	Identified gap
[13]	Fraud detection	AUC, precision	AdaBoost	No statistical validation or XAI
[14]	Real-world datasets	Accuracy	Ensemble learning	Only accuracy used, no metric comparison or validation
[21]	Marketing datasets	F1, AUC	Evolutionary methods	No comparative analysis or statistical validation
[18]	Business performance	Precision, recall	Decision Trees	Ignored MCC/F1, no statistical analysis
[19]	Financial datasets	Precision	Neural networks	Precision only; no comparison, no validation
[15]	MBA student data	Accuracy	Ensemble learning	Accuracy only; no metric selection analysis
[22]	Banking datasets	MCC	Complexity-based classifiers	No XAI or comparative metric analysis
[23]	Financial datasets	MCC	GANs	Only MCC; no validation or explainability
[17]	Insurance datasets	F1	AdaBoost	No statistical validation
[16]	Financial transactions	Accuracy	Spark + SVM	Accuracy only; ignores threshold/XAI
[20]	Bankruptcy datasets	Recall	SVM	Recall only; no comparative or XAI analysis
[24]	Credit scoring	F1	RNS classifiers	No statistical or XAI validation
[25]	Churn prediction	MCC	Ensemble models	No comparison or explainability
[28]	Business datasets	AUC	Boosting models	No metric selection or validation
[26]	Churn prediction	F1, G-Mean	Hybrid ensemble	No comparative metric analysis; no XAI
[27]	Churn evaluation	MCC, AUC	Multiple classifiers	No statistical methods or XAI applied

Table 2 Recent studies on metric evaluation and XAI under imbalanced learning

Ref.	Domain	Metric focus	XAI?	Stat.?	Identified gap
[9]	General	Accuracy, F1, MCC	✗	✗	Conceptual, not empirical comparison
[10]	General	Accuracy, F1, ROC, MCC	✗	✓	Compared metrics across domains; lacks business/XAI focus
[11]	General	Sampling, recall	✗	✗	Sampling-focused, not metric evaluation
[12]	Health data	Balanced Acc., MCC, F1	✗	✓	No XAI or threshold variation
[31]	General	ROC	✗	✗	Lacks imbalance-specific insight
[32]	General	MCC, ROC	✗	✓	Focused on bias correction, not metric interaction
[29]	Software defects	MCC, G-Mean	✗	✓	No threshold or explainability analysis
[30]	General	12 metrics incl. MCC, Kappa	✗	✓	Lacks XAI and threshold-dependent behavior
[37]	General	AUC, F-measure, G-mean, MCC	✗	✓	Theoretical consistency only; no XAI or dynamic threshold
[33]	Electricity demand	SHAP + Feature selection	✓	✗	Lacks metric integration
[34]	Time series	Metric for SHAP validation	✓	✗	Does not compare traditional metrics
[35]	Healthcare	SHAP + Risk prediction	✓	✗	SHAP only; no metric linkage
[36]	General	Taxonomy of XAI metrics	✓	✗	Review only; lacks experimental grounding

datasets—Default of Credit Card Clients and Telco Customer Churn—using a comprehensive methodology that includes ANOVA, McNemar’s test, bootstrap confidence intervals, threshold sensitivity analysis, noise robustness testing, and a novel two-stage SHAP-based XAI approach. The first stage applies conventional SHAP visualizations (bar plots and beeswarm plots) for baseline interpretability using LR, while the second

introduces 3D metric-conditioned SHAP analysis to explore how feature contributions vary across thresholds and relate to performance metrics. This is the first study to jointly analyze performance metrics across statistical, algorithmic, and interpretability dimensions, offering a generalizable and practitioner-oriented framework for metric selection in imbalanced business classification tasks.

Methodology

In this section the study presents the methodology of our investigation. The methodology of our study is designed to systematically evaluate and select the optimal performance metric for machine learning models applied to business data mining, with the focus on a large imbalanced Business datasets. To conduct a detailed analysis this research adopts a multi-faceted analytical framework integrating machine learning, advanced statistical techniques, and explainable AI (XAI). Figure 1 presents the research framework of our investigation containing all the phases adopted in this analysis.

Data preparation and preprocessing

The dataset used in this study is the “Default of Credit Card Clients” dataset collected from UCI Machine Learning, available at the URL¹, which consists of 30,000 instances and 23 features. It contains data from a Taiwanese bank’s credit card clients, specifically used to predict whether clients will default on their credit card payments. The features include demographic information, credit-related variables and payment amounts for six months, and the binary target variable indicating whether a client defaulted in the following month. Specifically, the target variable is denoted as DEFAULT_PAYMENT_NEXT_MONTH, where 1 represents default and 0 indicates no default. The features are

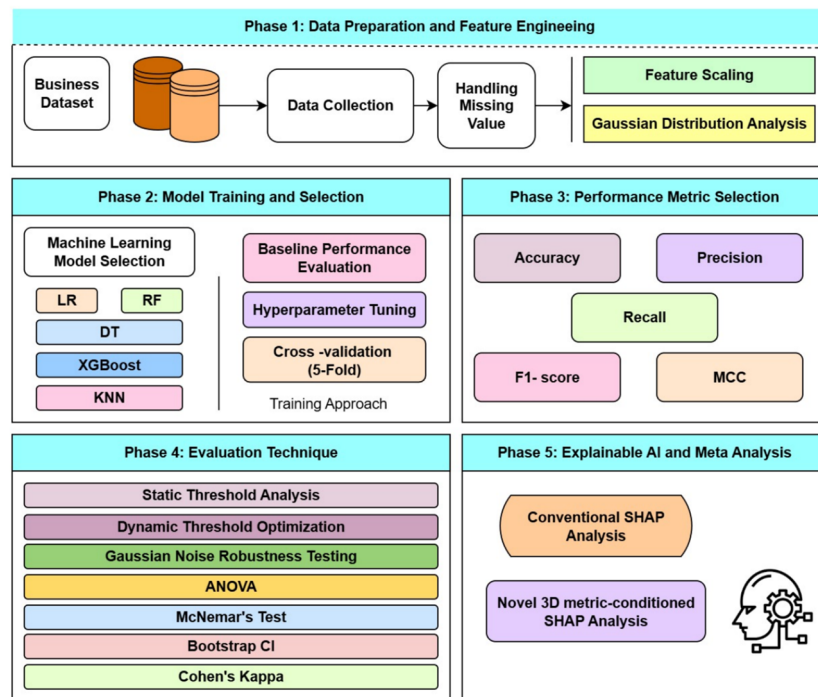
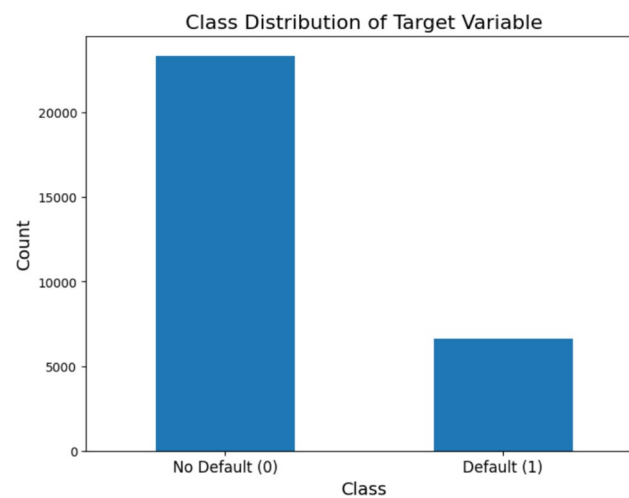


Fig. 1 Proposed research framework for selecting the best metric

¹ <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

Table 3 Feature summary and descriptions

Feature Name	Type	Note
ID, SEX, EDUCATION, MARRIAGE	Categorical	ID : Unique identifier for each client. SEX : Gender of the client (Male/Female). EDUCATION : Education level (1: Graduate school, 2: University, 3: High school, 4: Others). MARRIAGE : Marital status (1: Married, 2: Single, 3: Others).
LIMIT_BAL, AGE	Numeric	LIMIT_BAL : Credit limit in the credit card account. AGE : Age of the client.
PAY_0 to PAY_6	Categorical	PAY_0 to PAY_6 : Payment status for the last 6 months, where 0 means no delay and 1–9 indicates the number of months the payment was delayed. (Months: September 2005 to February 2005).
BILL_AMT1 to BILL_AMT6	Numeric	BILL_AMT1 to BILL_AMT6 : Bill statement amounts for the last 6 months (Months: September 2005 to February 2005).
PAY_AMT1 to PAY_AMT6	Numeric	PAY_AMT1 to PAY_AMT6 : Amount paid for the last 6 months (Months: September 2005 to February 2005).
DEFAULT_PAYMENT_NEXT_MONTH	Categorical	DEFAULT_PAYMENT_NEXT_MONTH : Whether the client defaulted on the payment in the next month (1: Default, 0: No default).

**Fig. 2** Class distribution with the imbalanced dataset

a mix of categorical and numerical variables. We provide a summary of the features and their respective types in Table 3. The distribution of the target variable of the dataset is presented in Figure 2, which demonstrates the imbalanced nature of the dataset. The dataset was initially examined for missing values, inconsistencies, and anomalies. Features with missing values were imputed to ensure the integrity of our selected dataset. For categorical features such as education and marital status, we converted them into numerical features using the hot encoder method.

Additional dataset

To further enhance the generalizability of our findings, we extended our experiments to another dataset. The second dataset employed in this study is the Telco Customer Churn dataset, sourced from Kaggle and available at the URL². This dataset comprises 7,043 instances and more than 20 features, collected from a telecom company's customer database, and is used to predict customer churn—the likelihood that a customer will cancel their subscription to the service. The target variable in this dataset is Churn,

²<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

a binary indicator where ‘Yes’ signifies a customer who has churned, and ‘No’ represents a customer who has remained with the company. The dataset contains a mix of numerical and categorical features, including customer demographics (e.g., age, gender), account details (e.g., tenure, monthly charges), service usage patterns (e.g., internet service, online security), and payment methods (e.g., electronic billing). This dataset is well-suited for exploring imbalanced classification problems, with approximately 26.5% of customers having churned, which allows for a rigorous evaluation of different performance metrics tailored to business data mining applications.

Feature scaling

Since the dataset contained various features with different units and ranges, all numerical features were standardized using z-score normalization. The study used standardization because it helps to increase the performance of business predictive models while dealing with large business datasets [38].

$$X' = \frac{X - \mu}{\sigma} \quad (1)$$

where X is the original feature value, μ is the mean, and σ is the standard deviation. This ensures that models sensitive to feature magnitudes and perform effectively.

Dataset splitting

The data was split into training (80%) and testing (20%) subsets using stratified sampling to maintain the class imbalance ratio in both sets. Stratified sampling ensures that the minority class (clients who defaulted) is represented proportionally, which is crucial for reliable model evaluation.

Model selection and validation of use

In this study, we aimed to evaluate the optimal performance metric for business data mining on imbalanced datasets. To ensure the generalizability and robustness of our findings on performance metric evaluation, we selected five machine learning models representing diverse algorithmic paradigms: LR as a linear model, DT, RF, and XGB as tree-based models, and KNN as an instance-based method. This variety enables a comprehensive assessment of metric behavior across fundamentally different learning strategies commonly used in business data mining.

Logistic Regression (LR)

LR is a linear model widely used for binary classification tasks. It is interpretable and computationally efficient, making it ideal for applications such as credit scoring and fraud detection in business contexts [39]. Its probabilistic output makes it suitable for threshold-based analyses. The decision boundary for LR is modeled as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}} \quad (2)$$

where β_0 is the intercept and $\beta_1, \dots, \beta_n$ are the coefficients for the features $X_1, \dots, X_n$. Regularization was applied to prevent overfitting, and hyperparameters were optimized using grid searching technique.

Random Forest (RF)

RF is an ensemble learning method that builds multiple DTs and averages their outputs to improve robustness and reduce overfitting [40]. We included the RF model due to its capability to manage high-dimensional datasets and capture complex interactions between features. Additionally, it is robust to class imbalances and can rank features' importance, aligning well with explainability needs in business domains. The prediction for RF is given by:

$$\hat{y} = \text{mode}\{h_1(X), h_2(X), \dots, h_T(X)\} \quad (3)$$

where $h_t(X)$ is the prediction of the t -th tree. The number of trees (T) and maximum tree depth were fine-tuned to achieve optimal performance.

Decision Tree (DT)

DT is an interpretable model that splits the dataset into subsets based on feature thresholds. Our study used this model, as it offers a simple yet interpretable structure for classification problems [41]. Their tree-like representation makes them easy to understand, even for non-technical stakeholders. The splitting criterion is determined using metrics such as Gini impurity:

$$G = 1 - \sum_{i=1}^C p_i^2 \quad (4)$$

where p_i is the proportion of instances belonging to class i , and C is the total number of classes. Pruning was applied to prevent overfitting.

XGBoost (XGB)

XGB is a gradient-boosting algorithm that optimizes a regularized objective function. It is able to prevent overfitting, making it particularly effective for datasets like ours with diverse feature distributions [42]. XGB's scalability and superior predictive performance make it highly applicable in business data mining. For binary classification, the objective function is defined as:

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{j=1}^T \Omega(h_j) \quad (5)$$

where ℓ is the loss function (log-loss for binary classification), and $\Omega(h_j)$ is the regularization term. Hyperparameters such as learning rate and tree depth were tuned using cross-validation.

k-Nearest Neighbors (KNN)

KNN is a distance-based algorithm that classifies instances based on the majority class among k nearest neighbors. We included KNN to analyze how non-parametric models, which rely on local decision boundaries, perform compared to parametric and ensemble models [43]. Its sensitivity to feature scaling allowed us to evaluate the impact of preprocessing steps on metric performance.

The classification is determined as:

$$\hat{y} = \operatorname{argmax}_c \sum_{i=1}^k I(y_i = c) \quad (6)$$

where I is an indicator function that equals 1 if $y_i = c$ and 0 otherwise. The value of k and the distance metric (Euclidean distance) were optimized for best performance.

In this research, the study used multiple machine learning models and statistical tests to assess their performance in selecting the best performance metrics. Table 4 presents the configuration of each model, including their specific hyperparameters and key settings. In addition, we outline the statistical tests and performance metrics used to evaluate the models we selected in this research. We provide a clear overview of the experimental setup for reproducibility and understanding of the analysis process that we incorporated in this investigation.

Performance metric evaluation

The evaluation of machine learning models in this study focused on five key performance metrics encompassing the Accuracy, Precision, Recall, MCC, and F1-Score. These metrics were chosen based on their relevance and robustness for imbalanced datasets and their ability to capture different dimensions of model performance. This section discusses the significance of each metric, their mathematical formulation, and why they were specifically chosen over other potential metrics.

Table 4 Parameter settings for selected models and techniques

Model/analysis	Parameter	Setting
Logistic regression	Max_iter	1000
	Solver	'lbfgs'
	Penalty	'l2'
	C	1.0
	Random_state	42
Random forest	N_estimators	100
	Max_depth	6
	Random_state	42
Decision tree	Max_depth	5
	Random_state	42
XG boost	n_estimators	100
	Learning_rate	0.1
	Max_depth	6
	Subsample	0.8
K-Nearest neighbors	n_neighbors	5
	Weights	'uniform'
Cross-validation	cv	5
	Scoring	['accuracy','precision','recall','f1','mcc']
SHAP analysis	Explainer	shap.Explainer(model, X_train)
	Shap_values	explainer(X_test)
	Mean SHAP importance	Absolute mean SHAP values per feature
Statistical analysis	Bootstrap Confidence Intervals	1000 bootstraps
	McNemar's Test	Exact p-value
	Cohen's Kappa	Measure agreement between model
	ANOVA	Testing significant differences across metrics
	Significance Level	p-value < 0.05 for significance

Accuracy

While it is widely recognized that accuracy is a poor standalone metric in imbalanced classification settings—often yielding deceptively high values by favoring the majority class [9, 44]—its inclusion in this study serves a critical analytical role. Rather than using accuracy as a primary indicator of model performance, we treat it as a control metric or baseline to highlight the limitations of naive evaluation in the presence of imbalance. By systematically comparing accuracy to more robust alternatives (e.g., F1-score, MCC) across varying imbalance ratios and classification thresholds, we expose the instability and over-optimism of accuracy in high-stakes business domains such as churn and credit default prediction. Furthermore, the divergence between accuracy and minority-sensitive metrics under conditions of noise and threshold shifts is quantitatively analyzed using bootstrap confidence intervals and ANOVA, reinforcing the need for multi-metric evaluation. This comparative lens allows us not only to validate the inadequacy of accuracy empirically but also to showcase its misalignment with business-relevant decision-making. Accuracy is mathematically expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives.

Precision

Precision quantifies the proportion of positive predictions that are correct:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

Precision has been included in our investigation because it plays a crucial role in business applications where false positives have high costs, such as fraud detection or customer churn prediction. It ensures that the model does not excessively classify instances as positive when they are not, which is highly relevant for business data mining [45].

Recall

Recall mainly refers to the percentage of actual positive cases that are correctly identified by the specific model:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

Recall is vital when false negatives are costly, such as missing a default customer in credit scoring. It ensures the model captures as many true positives as possible, which is essential for risk management in business data mining [46].

F1-score

The F1-Score is the harmonic mean of Precision and Recall, balancing the trade-off between the two:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

The F1-Score offers a single metric that balances precision and recall and provides equal importance to both, making it particularly useful for imbalanced datasets. It prevents over-optimization of one metric at the expense of the other [47].

Matthews Correlation Coefficient (MCC)

MCC is a robust metric that considers all four confusion matrix elements (TP , TN , FP , FN) to provide a balanced evaluation:

$$MCC = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (11)$$

MCC is ideal for imbalanced datasets as it accounts for true and false classifications of both classes. Unlike Accuracy, it remains reliable even when class distributions are highly skewed [48].

Evaluation of additional metrics

In addition to our selected five primary evaluation metrics, including Accuracy, Precision, Recall, F1-score, and MCC, we included four alternative metrics commonly used in imbalanced classification problems, such as Geometric Mean (G-Mean), Area Under the ROC Curve (AUC), Balanced Accuracy, and the Index of Balanced Accuracy (IBA). These metrics were computed using the same prediction outputs from the trained models at a fixed decision threshold of 0.5. Their inclusion aims to offer a broader perspective on model performance and to respond to established critiques in the literature regarding the limitations of conventional metrics in imbalanced scenarios. However, these alternative metrics were not included in subsequent robustness, statistical testing, or explainability analysis. This decision was based on their performance consistency with our selected five metrics during baseline comparison, as well as their lower interpretability in business decision-making contexts. As such, the remainder of the evaluation focuses on the five core metrics.

Cost-sensitive evaluation

To assess how well our selected conventional evaluation metrics align with real-world financial decision-making, we implemented a cost-sensitive evaluation framework based on confusion matrix outputs. This approach simulates domain-specific misclassification penalties by applying a cost matrix where false positives incur a cost of 10 units, and false negatives incur a cost of 100 units—reflecting business-critical contexts such as loan default or customer churn. For each model, we computed the Expected Cost of Misclassification (ECM) using the formula:

$$ECM = (FP \times C_{FP}) + (FN \times C_{FN}) \quad (12)$$

where $C_{FP} = 10$ and $C_{FN} = 100$. We also derived a Net Profit score by taking the inverse of the ECM. These results were used to evaluate the practical alignment of F1-score and MCC with cost-based priorities.

Our design of this cost-sensitive evaluation is motivated by prior work in financial domains, where researchers have highlighted the limitations of accuracy-driven metrics and proposed profit-oriented alternatives. For example, rank-based portfolio selection approaches have been shown to outperform market indices by explicitly linking

evaluation to investment returns [49]. Similarly, cost-sensitive prediction frameworks have been applied to stock price forecasting, where tailored misclassification costs and feature selection improved investment outcomes [50]. Inspired by these studies, we integrate cost-adjusted metrics into our evaluation pipeline to better reflect financial and business realities.

Stratified 5-fold cross-validation

In this section, we implemented 5-fold stratified Cross-Validation as presented in Algorithm 1 to evaluate multiple models on our selected large business dataset, consisting of 30,000 instances. This approach balances computational efficiency with reliable performance estimation by splitting the data into 5 folds, using each fold as a test set while training on the remaining data. The process is repeated across all folds, and for each model, we calculate the mean and standard deviation of the performance metrics—accuracy, precision, recall, F1-score, and MCC. This allows us to assess both the central tendency and variability of model performance. The 5-fold cross-validation ensures that each model is evaluated across different data subsets, providing robust estimates of performance and helping to select the most appropriate metric for model comparison. This approach is particularly effective for large datasets like ours, as it mitigates the risk of overfitting while offering a comprehensive assessment of each model's reliability in a business data mining scenario [51]. Our proposed Stratified 5-Fold Cross-Validation process has been presented in Figure 3 meticulously.

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1: Input: Training Data  $D = \{X_{\text{train}}, y_{\text{train}}\}$ , Models  $M = \{M_1, M_2, \dots, M_m\}$ , Number of folds  $k = 5$ , Evaluation metrics  $\mathcal{M} = \{\text{accuracy, precision, recall, F1, MCC}\}$ 
2: Output: Mean and standard deviation of performance metrics for each model
3: for each model  $M_i \in M$  do
4:   Initialize an empty list to store performance metrics for each fold
5:   Split the dataset  $D$  into  $k = 5$  stratified folds
6:   for each fold  $j = 1, 2, \dots, 5$  do
7:     Let  $D_{\text{train}}^{(j)}$  be the training set (all data except fold  $j$ )
8:     Let  $D_{\text{test}}^{(j)}$  be the test set (fold  $j$ )
9:     Train model  $M_i$  on  $D_{\text{train}}^{(j)}$ 
10:    Predict  $\hat{y}_{\text{test}}^{(j)}$  on  $D_{\text{test}}^{(j)}$ 
11:    for each metric  $m \in \mathcal{M}$  do
12:      Compute  $m_j$  for metric  $m$  on the test set
13:    end for
14:  end for
15:  Compute the mean and standard deviation for each metric  $m \in \mathcal{M}$ :
16:   $\mu_m(M_i) = \frac{1}{5} \sum_{j=1}^5 m_j$ 
17:   $\sigma_m(M_i) = \sqrt{\frac{1}{5} \sum_{j=1}^5 (m_j - \mu_m(M_i))^2}$ 
18: end for
19: Store the results for each model in a table with mean and standard deviation for each metric

```

Algorithm 1 5-Fold Stratified Cross-Validation for Model Evaluation

Statistical validation

In this study we implemented four statistical tests, including ANOVA, bootstrap confidence intervals, McNemar's test, and Cohen's kappa. The combination of these statistical techniques ensures that the selection of the optimal metric is not based solely on

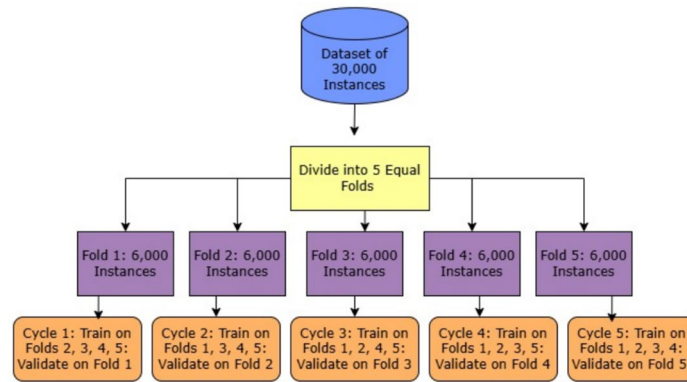


Fig. 3 Proposed stratified 5-fold cross-validation

numerical differences but is statistically validated. By identifying significant differences and quantifying the reliability of the metric, this study ensures that the selected metric, whether F1-score, MCC, or another, provides the most robust and generalizable assessment of the model performance for business data mining tasks. These methods directly address the gaps in current research, where statistical validation is often overlooked in metric selection.

Analysis of variance (ANOVA)

To directly evaluate the relative reliability of performance metrics across models and thresholds, we applied a repeated-measures analysis of variance (ANOVA). This approach tests for significant differences between metrics while accounting for the repeated structure of the data (i.e., metric scores across multiple models and thresholds).

For each of our selected metrics, including Accuracy, Precision, Recall, F1-score, and MCC, we collected scores across five classifiers and nine thresholds ($0.1 \leq \tau \leq 0.9$). We then performed a within-subjects repeated-measures ANOVA with Metric as the within factor. Formally, the ANOVA model can be expressed as:

$$Y_{ij} = \mu + \alpha_i + s_j + \epsilon_{ij} \quad (13)$$

where Y_{ij} is the performance score of metric i for subject (model-threshold combination) j , μ is the grand mean, α_i is the effect of metric i , s_j is the random effect of subject j , and ϵ_{ij} is the residual error term. The null hypothesis tested was:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k \quad (14)$$

where $k = 5$ metrics, against the alternative that at least one metric mean differs. Where significant main effects were observed ($p < 0.05$), we conducted Holm-corrected pairwise post-hoc comparisons. For each pair of metrics a and b , the test statistic was computed as:

$$t = \frac{\bar{X}_a - \bar{X}_b}{SE(\bar{X}_a - \bar{X}_b)} \quad (15)$$

with adjusted p -values using the Holm-Bonferroni method to control the family-wise error rate. To complement the inferential tests, we also computed descriptive statistics

for each metric, including the mean, median, standard deviation (σ), interquartile range (IQR), and coefficient of variation (CV):

$$CV = \frac{\sigma}{\mu} \quad (16)$$

Boxplots were generated to visualize the distribution and stability of metric scores across all models and thresholds.

McNemar's test

We use McNemar's test to compare prediction errors between two models [52]. This test determines if there is a statistically significant difference in the classification errors of two models on the same dataset, making it especially relevant for metrics like precision and recall. It highlights differences in error distributions, which are critical for understanding trade-offs between precision and recall. The test statistic is calculated as:

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad (17)$$

where:

- b is the count of instances misclassified by model A but correctly classified by model B,
- c is the count of instances correctly classified by model A but misclassified by model B.

This test highlights differences in the models' classification capabilities, particularly for imbalanced datasets, where error patterns are critical for evaluating performance metrics.

Bootstrap confidence intervals

Bootstrap confidence intervals (CIs) were calculated to quantify the variability and reliability of each performance metric. It provides insights into metric variability, ensuring the reliability of selected metrics under different scenarios [53]. By resampling the data with replacement N times ($N = 1000$), we obtained a distribution of metric values and computed the 95% confidence interval as:

$$CI_{95\%} = [P_{2.5\%}, P_{97.5\%}] \quad (18)$$

where, $P_{2.5\%}$ and $P_{97.5\%}$ are the 2.5th and 97.5th percentiles of the bootstrap distribution. Metrics with narrower confidence intervals were deemed more reliable, providing robust guidance for metric selection in noisy and imbalanced settings.

Cohen's kappa

We used Cohen's Kappa (κ) to measure the agreement between two models' predictions, accounting for the agreement occurring by chance [54]. It is defined as:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (19)$$

where:

- P_o is the observed agreement between the two models,
- P_e is the expected agreement due to chance.

Cohen's Kappa provides a scale of agreement, where:

- $\kappa > 0.8$: strong agreement,
- $0.6 < \kappa \leq 0.8$: moderate agreement,
- $\kappa \leq 0.6$: weak agreement.

This metric is particularly useful for validating classification consistency across models and evaluating metrics such as F1-Score and MCC, which are sensitive to class imbalance.

Threshold sensitivity analysis

Static threshold sensitivity analysis

Static threshold analysis involves evaluating model performance metrics: accuracy, precision, recall, F1-score, and MCC across fixed decision thresholds, ranging from 0.1 to 0.9. The goal is to understand the trade-offs between metrics and their behavior as thresholds change [55]. The performance metrics for a given threshold t are calculated based on predictions $\hat{y}$, defined as:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1 | X) \geq t, \\ 0 & \text{if } P(y = 1 | X) < t. \end{cases}$$

Dynamic threshold sensitivity analysis

Dynamic threshold analysis optimizes thresholds for each metric to maximize performance. For each metric M , the optimal threshold t^* is determined as:

$$t^* = \arg \max_t M(t) \quad (20)$$

where $M(t)$ is the value of the metric at the threshold t .

This analysis identifies the best threshold for accuracy, precision, recall, F1-score, and MCC individually, offering tailored insights into the performance of metric-specific models. This analysis helps identify decision thresholds where specific metrics reach their peak performance. In addition, it provides actionable recommendations for business decision-making, where balancing false positives and false negatives is crucial.

Noise robustness testing

Noise robustness testing evaluates how performance metrics behave under varying noise levels in feature inputs. Gaussian noise with standard deviation σ is added to the input features X to simulate real-world data inconsistencies [56]. The noisy feature X_{noisy} is defined as:

$$X_{\text{noisy}} = X + N(0, \sigma^2) \quad (21)$$

We trained our selected models on the original dataset and tested on noisy data, the range of the noise levels was from $\sigma = 0.0$ (no noise) to $\sigma = 0.3$. It evaluates metric stability under noisy conditions, reflecting real-world scenarios where data may contain

errors. Finally, it identifies models resistant to noise, crucial for business data mining, where data quality often varies based on the scenario.

Explainable AI (XAI) analysis

To improve interpretability for business end-users, we employed SHAP (SHapley Additive exPlanations), a widely used explainability technique based on cooperative game theory. SHAP decomposes a model's output $f(x)$ into additive feature contributions:

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i \quad (22)$$

Here, ϕ_0 is the expected model output, and ϕ_i represents the marginal contribution of feature x_i to the prediction. We applied SHAP to an LR model using the LinearExplainer, with a representative background sample from the training set to ensure stability. We selected the LR model due to its transparency and compatibility with additive feature attribution methods, making it ideal for explainable business applications. Our analysis follows a two-stage framework. In the first stage, we generated conventional SHAP outputs, including summary bar plots and beeswarm plots, which highlight both global and local feature importance. These plots illustrate the average and instance-specific influence of each feature on the model's predictions.

In the second stage, we introduce a novel 3D SHAP framework to link model interpretability with metric-specific performance across varying classification thresholds. Thresholds were varied from 0.1 to 0.9, and for each value, classification outcomes were generated on the test set. Evaluation metrics were computed alongside SHAP values. To quantify feature importance at each threshold, we computed the mean absolute SHAP value per feature across all test samples, defined as:

$$S_i(t) = \frac{1}{n} \sum_{j=1}^n \left| \phi_{i,j}^{(t)} \right| \quad (23)$$

where $S_i(t)$ is the metric-conditioned SHAP importance of feature x_i at threshold t , $\phi_{i,j}^{(t)}$ is the SHAP value for feature x_i and sample j at threshold t , and n is the number of test instances.

This enabled the construction of 3D visualizations in which the X-axis represents the classification threshold, the Y-axis reflects SHAP importance, and the Z-axis corresponds to the value of a selected evaluation metric. These plots reveal how the influence of individual features shifts across different decision boundaries and performance criteria, offering deeper insight into metric behavior under realistic business constraints.

Finally, the combined use of static and dynamic threshold analysis, statistical analysis, noise robustness testing, and SHAP-based explainability creates a comprehensive framework for selecting the optimal performance metric in business data mining. The general experimental design has been described in Figure 4. To provide the detailed phases of our analysis, Algorithm 2 presents the overall steps that have been followed in this rigorous investigation. With the experimental setup in place, we now report the performance metrics and statistical outcomes derived from our evaluation.

-
- 1: **Input:** Dataset D, Models M, Metrics P, Thresholds T, Noise levels σ .
 - 2: **Output:** Optimal metric, model, and threshold.
 - 3: Data Preprocessing
 - 3.1 Standardize all numerical features to ensure consistent scaling and support models sensitive to feature magnitudes.
 - 4: Data Preparation
 - 4.1 Standardize all numerical features to ensure consistent scaling and support models sensitive to feature magnitudes.
 - 5: Model Training
 - 5.1 Train five machine learning models (Logistic Regression, Random Forest, Decision Tree, XGBoost, and k-Nearest Neighbors) using stratified 5-fold cross-validation.
 - 5.2 Compute average performance metrics across folds for each model to ensure fair evaluation.
 - 6: Threshold Sensitivity Analysis
 - 6.1 Static Analysis: Evaluate performance metrics for fixed thresholds (0.1 to 0.9) to analyze trade-offs and identify metrics that remain stable across thresholds.
 - 6.2 Dynamic Analysis: Optimize thresholds for each metric to determine the point where each metric achieves its highest value.
 - 7: Noise Robustness Testing
 - 7.1 Inject varying levels of Gaussian noise into feature inputs to simulate real-world data inconsistencies.
 - 7.2 Assess metric stability across noise levels (0.0 to 0.3) to identify models and metrics resilient to noise.
 - 8: Statistical Validation
 - 8.1 Repeated-measures ANOVA: Testing for significant differences across performance metrics.
 - 8.2 McNemar's Test: Compare error distributions between models to evaluate classification trade-offs.
 - 8.3 Cohen's Kappa: Assess the agreement between model predictions, accounting for chance agreement.
 - 8.4 Bootstrap Confidence Intervals: Quantify metric reliability by calculating confidence intervals for each metric across models.
 - 9: Explainable AI (XAI) Analysis
 - 9.1 Use SHAP values to interpret feature contributions to model predictions.
 - 9.2 Analyze how feature importance changes across thresholds and metrics to link model interpretability with performance.
 - 10: Metric and Model Selection
 - 10.1 Evaluate metrics based on stability across thresholds, robustness under noise, and statistical significance.
 - 10.2 Recommend the most reliable metric, optimal model, and corresponding threshold for business data mining tasks.
-

Algorithm 2 Framework for Optimal Performance Metric Selection

Experimental results

In this section we present the experimental results of our investigation. To evaluate the reliability of performance metrics in business data mining, we conducted extensive experiments on two real-world benchmark datasets: the large-scale Default of Credit Card Clients dataset and the smaller Telco Customer Churn dataset. These datasets differ in size and imbalance levels but are both representative of practical business prediction scenarios. For the first dataset, we applied a comprehensive suite of evaluations, including exploratory data analysis, cross-validation, static and dynamic threshold sensitivity analysis, noise robustness testing, statistical significance testing, and two-stage SHAP-based explainability.

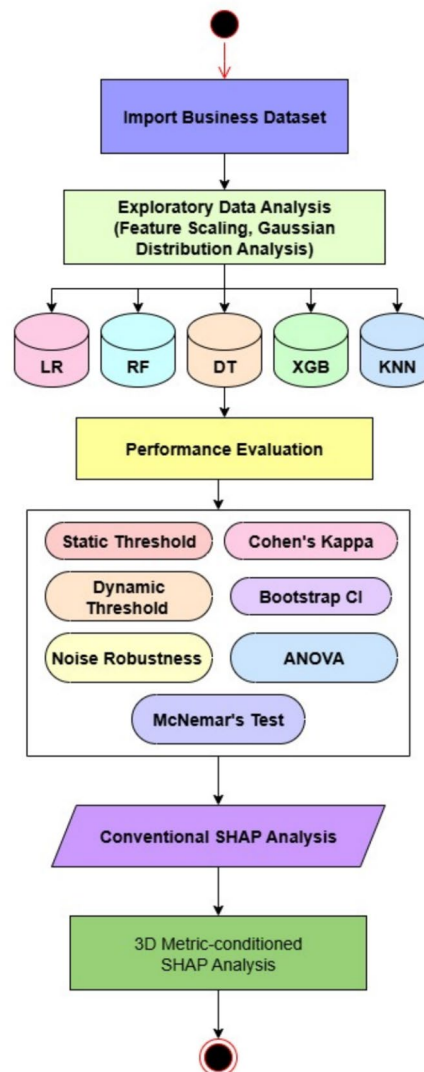


Fig. 4 Experimental design for selecting the best performance metrics

To ensure generalizability of our findings while considering the experimental length, we focused on key evaluations for the second dataset. Specifically, we conducted cross-validation, Gaussian noise robustness testing, and bootstrap confidence intervals and applied advanced statistical tools such as McNemar's test and Cohen's kappa to mathematically assess pairwise model agreement and classification significance. Additionally, we employed SHAP-based explainable AI techniques to analyze feature importance across performance metrics and classification thresholds. This structured and statistically rigorous approach ensures that our results are robust, interpretable, and applicable to diverse business data mining contexts.

Exploratory data analysis

To establish a robust foundation for analyzing the performance of our selected machine learning models, our study conducted an initial exploration of the dataset, focusing on the distributional properties of key features. To better understand the characteristics of the dataset, Gaussian distribution plots were generated for 12 selected features as

displayed in Figure 5a to Figure 5i. These plots provide critical insights into the statistical properties of each feature, including their central tendency, spread, and skewness, which are essential to validate the suitability of different machine learning models and performance metrics. Gaussian distribution analysis is particularly relevant, as many machine learning algorithms, such as LR, assume features are normally distributed, while others, such as tree-based models, are less sensitive to such assumptions. This analysis also helps identify features that may require scaling, transformation, or special handling [57], especially in the context of imbalanced distributions or outliers, which can impact metrics such as F1-Score and MCC that emphasize the balance between true positives and false negatives.

Key insights were derived from individual feature distributions. LIMIT_BAL in Figure 5a exhibited a strong right-skewed distribution, with most clients concentrated at lower credit limits, potentially leading to imbalances during training. Binary features such as SEX in Figure 5b displayed a bimodal distribution, confirming a balanced representation of demographic groups, while categorical features such as EDUCATION in Figure 5c and MARRIAGE in Figure 5d demonstrated multi-modal distributions reflecting the diversity of education levels and marital statuses. Similarly, AGE in Figure 5e showed a right-skewed distribution, with younger clients dominating the dataset and a long tail for older clients. Payment history features such as PAY_0, PAY_2, and PAY_3 in

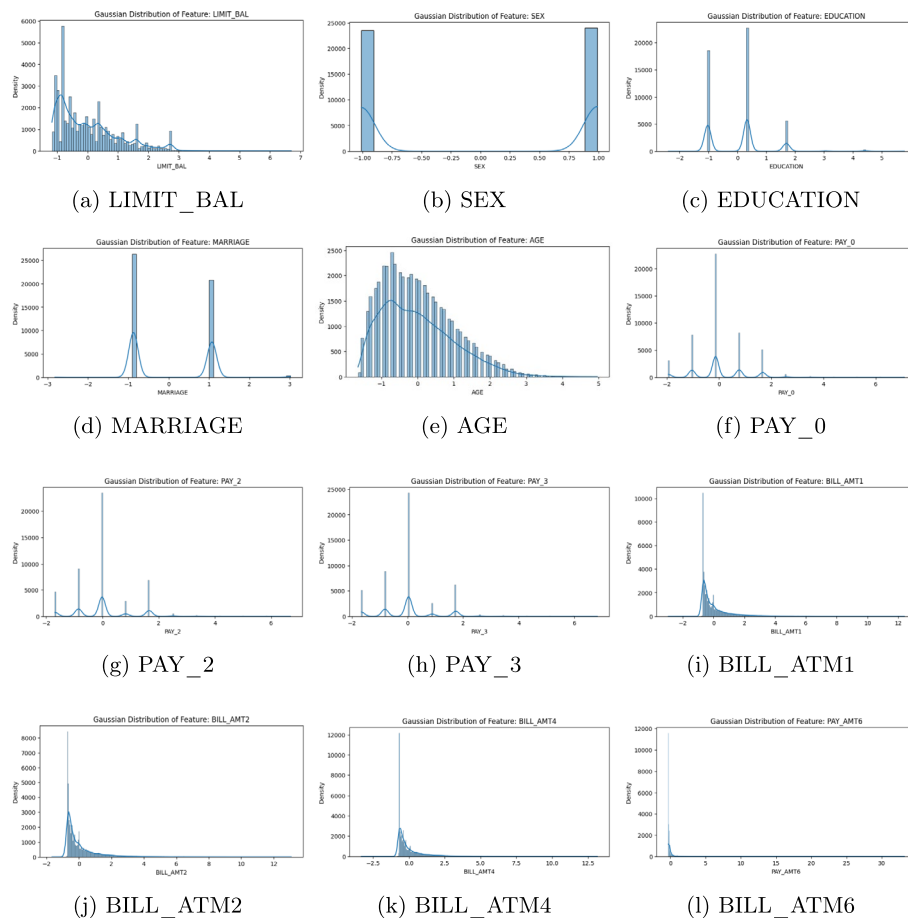


Fig. 5 Gaussian distribution of the Default of Credit Card Clients dataset for different features. The description of the feature labels is given in [58] (a) LIMIT_BAL, (b) SEX, (c) EDUCATION, (d) MARRIAGE, (e) AGE, (f) PAY_0, (g) PAY_2, (h) PAY_3, (i) BILL_ATM1, (j) BILL_ATM2, (k) BILL_ATM4, (l) BILL_ATM6

Figure 5f to Figure 5h revealed a significant peak at zero, corresponding to no delays in payments, while additional peaks indicated varying levels of delinquency, emphasizing the predictive importance of repayment behaviors.

Billing amounts BILL_AMT1, BILL_AMT2, and BILL_AMT4 in Figure 5i to Figure 5k displayed consistent right-skewed distributions, with most values clustered near zero and a long tail extending to higher billing amounts. This variability underscores the importance of scaling these features to avoid biasing distance-based models such as KNN. Similarly, PAY_AMT6 in Figure 5l exhibited an extreme right-skewed distribution, highlighting the need to account for rare but significant large payment values during model training. These findings emphasize the need for robust preprocessing techniques, including standardization and transformations, to ensure fair contributions from all features. Moreover, the analysis provides valuable insights into how feature distributions impact the suitability of performance metrics, with metrics such as MCC and F1-Score being particularly effective in handling imbalanced and skewed data distributions. This detailed understanding of the dataset lays the groundwork for the rigorous evaluation of machine learning models and performance metrics in subsequent sections.

Baseline performance evaluation

To establish a benchmark for performance metrics, we evaluated five machine learning models: LR, RF, DT, XGB, and KNN. The evaluation mainly focused on our selected five performance metrics. Table 5 provides a summary of the baseline performance metrics for each model, while Figure 6a to Figure 6e present bar plots to visually assess and compare the performance of the models based on these performance indicators. We observed that LR achieved the highest precision (0.657), indicating its effectiveness in minimizing false positives, although it struggled with recall (0.197), reflecting limited sensitivity to correctly identify defaults [59]. XGB demonstrated the most balanced performance, achieving the highest F1-Score (0.380) and MCC (0.287), making it robust in handling class imbalances [60]. DT, on the other hand, achieved the highest recall (0.404) but at the cost of low precision (0.346), resulting in a lower overall F1-Score and MCC. RF showed slightly better recall (0.238) compared to LR and KNN, but its performance was less consistent across all metrics. KNN delivered moderate performance, with an Accuracy of 0.772 and MCC of 0.212, reflecting its sensitivity to feature scaling and its limitations in handling skewed distributions [61].

These results highlight the trade-offs that we encountered across models. For instance, while LR excelled in precision, it missed many true positives due to its lower recall. In contrast, DT and XGB captured more true positives, as evidenced by their higher recall, but were more prone to false positives. Metrics such as F1-Score and MCC proved invaluable in balancing these trade-offs, helping us identify XGB as the most balanced baseline model. These findings underscore the critical importance of selecting

Table 5 Baseline model performance for each metric

Model	Accuracy	Precision	Recall	F1-Score	MCC
LR	0.800	0.657	0.197	0.303	0.280
RF	0.787	0.541	0.238	0.330	0.253
DT	0.700	0.346	0.404	0.373	0.178
XGB	0.790	0.546	0.291	0.380	0.287
KNN	0.772	0.469	0.238	0.315	0.212

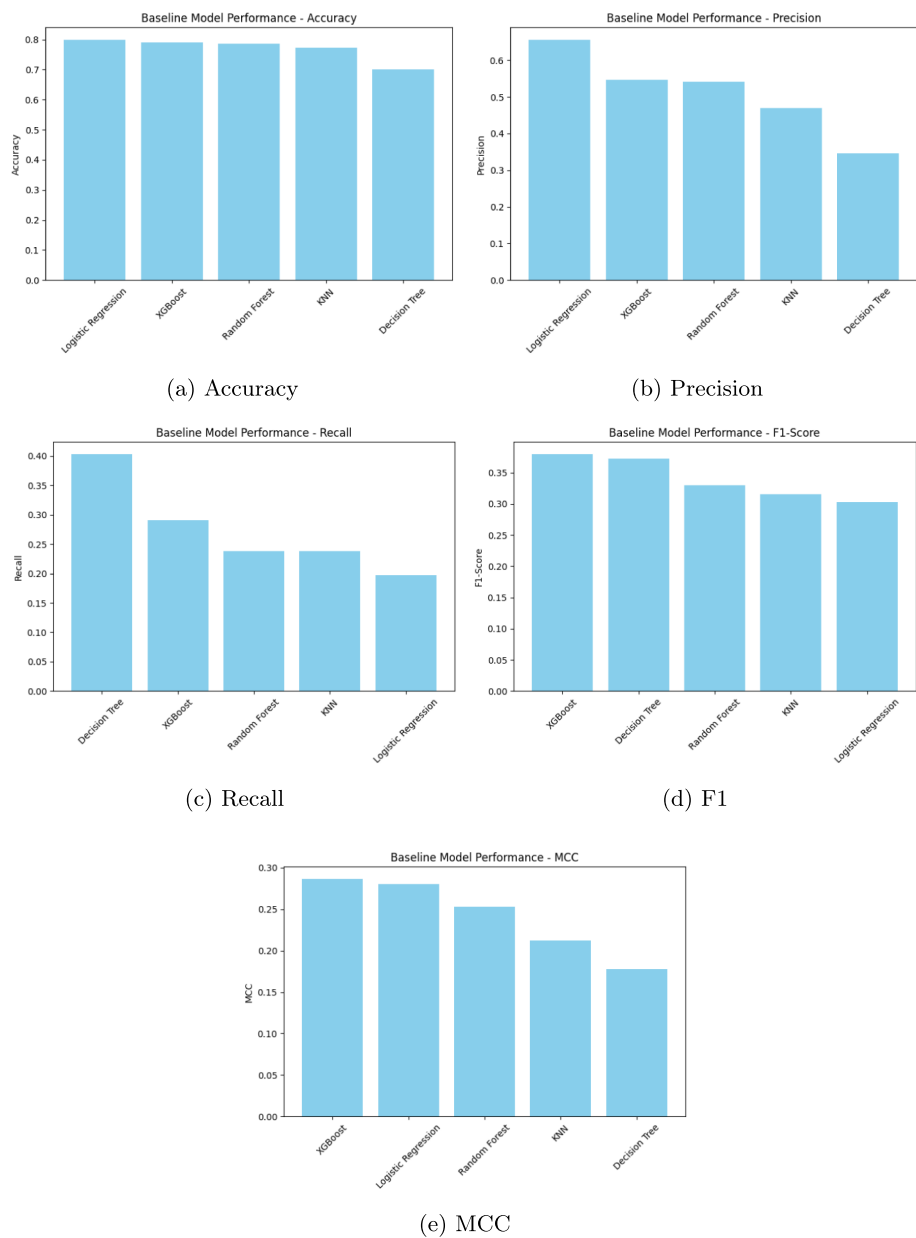


Fig. 6 Baseline performance evaluation for all models across metrics: **(a)** accuracy, **(b)** precision, **(c)** recall, **(d)** F1, and **(e)** MCC

appropriate metrics for evaluating models in business data mining, where specific goals, such as reducing false positives or capturing all true positives, may guide both metric and model selection.

Performance with cross validation

To evaluate the reliability and robustness of the models, we performed a 5-fold stratified cross-validation analysis across five metrics: Accuracy, Precision, Recall, F1-Score, and MCC. Table 6 summarizes the mean and standard deviation of each metric across the folds, while Figure 7a to Figure 7e present error bar plots that illustrate the variability in metric performance across folds. The error bar plots provided valuable insights into the consistency of the models. LR exhibited minimal variability in accuracy, with a mean of

0.801 and a standard deviation of 0.006, suggesting that it performs consistently across different folds. Precision for LR also displayed tight error bars, highlighting its reliability in minimizing false positives. RF and XGB exhibited slightly higher variability in their F1-Scores and MCC values, although their average performance remained robust, with RF achieving the highest MCC (0.313) and XGB obtaining the second-highest F1-Score (0.391). DT showed the highest variability in recall, reflecting its sensitivity to class imbalance and fold-specific characteristics, while KNN demonstrated the largest variability in F1-Score and MCC, indicating its reliance on consistent feature distributions across folds.

These results emphasize the importance of analyzing metric variability alongside mean performance. Models like LR and RF demonstrated not only high average performance but also low variability, making them suitable for business data mining tasks that demand reliability. In contrast, the higher variability observed for DT and KNN suggests that these models may require additional tuning or preprocessing to achieve consistent results in practice. The error bar analysis reinforces the importance of cross-validation as a tool for assessing model robustness and consistency, ensuring their suitability for real-world business applications.

Static threshold sensitivity analysis

To evaluate the impact of varying static classification thresholds on model performance, we analyzed five key metrics—Accuracy, Precision, Recall, F1-Score, and MCC—across a range of thresholds (0.1 to 0.9). The goal was to identify the trade-offs inherent in threshold adjustments and their effects on model performance, particularly for business data mining tasks where balancing Precision and Recall is critical. Table 7 summarizes the performance metrics for LR, RF, DT, XGB, and KNN models across thresholds. Figure 8a to Figure 8e visualize the threshold-performance relationships in three-dimensional plots, where the x-axis denotes the threshold, the y-axis denotes the metric score, and the z-axis corresponds to the model.

The results demonstrate notable trade-offs between Precision and Recall as thresholds change. For example, Precision improves significantly for most models as the threshold increases, reflecting a reduction in false positives. In contrast, recall decreases sharply at higher thresholds as a result of stricter classification criteria, leading to an increased rate of false negatives [62]. This is evident in models such as LR and RF, which achieve high precision but suffer in Recall at thresholds above 0.5. Metrics like Accuracy and MCC exhibit more stability across intermediate thresholds, particularly for models like XGB and LR, suggesting their robustness in scenarios requiring a balance between false positives and false negatives. The three-dimensional plots further highlight the trends across models. For instance, in Figure 8a, Accuracy increases rapidly at lower thresholds and stabilizes around intermediate values for most models, with LR and XGB showing

Table 6 Results with 5-fold stratified cross-validation

Model	Mean Accuracy	Std Accuracy	Mean Precision	Std Precision	Mean Recall	Std Recall	Mean F1	Std F1	Mean MCC	Std MCC
LR	0.801	0.006	0.688	0.064	0.188	0.021	0.294	0.028	0.284	0.031
RF	0.799	0.006	0.597	0.022	0.288	0.029	0.388	0.030	0.313	0.027
DT	0.716	0.015	0.370	0.028	0.399	0.035	0.384	0.028	0.200	0.036
XGB	0.785	0.009	0.525	0.032	0.312	0.030	0.391	0.028	0.285	0.030
KNN	0.779	0.008	0.498	0.033	0.267	0.036	0.347	0.038	0.245	0.038

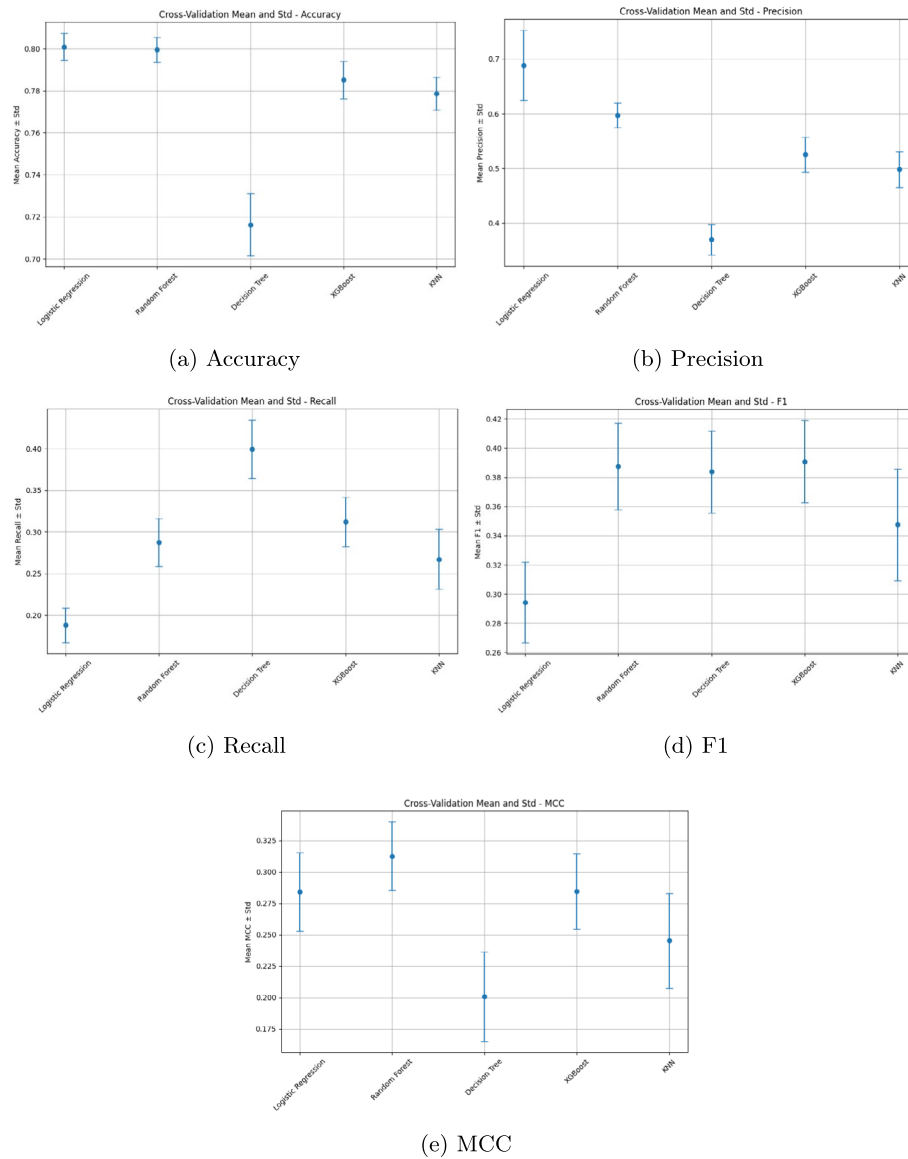


Fig. 7 Cross-validation mean and standard deviation for all models across metrics: (a) accuracy, (b) precision, (c) recall, (d) F1, and (e) MCC

the most consistent performance. Figure 8b and Figure 8c reveal opposing trends for Precision and Recall, while Figure 8d shows that F1-Score, which balances Precision and Recall, reaches optimal levels at thresholds of 0.4 to 0.6. Similarly, Figure 8e indicates that MCC, which accounts for both true and false predictions, peaks around these intermediate thresholds. These results emphasize the need for careful threshold tuning, particularly in business applications where trade-offs between Precision and Recall must align with specific operational objectives. Through this analysis, we underline the importance of selecting thresholds that maximize overall performance while accounting for the priorities of the business context.

Dynamic threshold sensitivity analysis

In this section we conducted Dynamic threshold sensitivity analysis to identify the optimal thresholds for maximizing key performance metrics—Accuracy, Precision, Recall,

Table 7 Static threshold sensitivity analysis across different metrics

Model & Metric	Threshold								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
LR Accuracy	0.340	0.574	0.767	0.793	0.800	0.791	0.782	0.782	0.778
LR Precision	0.241	0.287	0.468	0.567	0.657	0.700	0.588	0.667	0.400
LR Recall	0.924	0.623	0.395	0.265	0.197	0.094	0.045	0.027	0.009
LR F1	0.382	0.393	0.428	0.361	0.303	0.166	0.083	0.052	0.018
LR MCC	0.113	0.153	0.285	0.283	0.280	0.202	0.116	0.102	0.031
RF Accuracy	0.412	0.610	0.745	0.782	0.787	0.784	0.785	0.782	0.780
RF Precision	0.254	0.310	0.429	0.509	0.539	0.547	0.625	0.636	1.000
RF Recall	0.857	0.623	0.475	0.377	0.251	0.130	0.067	0.031	0.005
RF F1	0.391	0.414	0.451	0.433	0.343	0.210	0.122	0.060	0.009
RF MCC	0.211	0.222	0.254	0.279	0.280	0.252	0.259	0.226	0.184
XGB Accuracy	0.624	0.708	0.754	0.779	0.790	0.789	0.795	0.792	0.788
XGB Precision	0.318	0.371	0.434	0.500	0.546	0.556	0.614	0.628	0.636
XGB Recall	0.614	0.462	0.386	0.332	0.292	0.224	0.193	0.144	0.094
XGB F1	0.419	0.411	0.409	0.399	0.380	0.320	0.294	0.234	0.164
XGB MCC	0.202	0.222	0.254	0.279	0.287	0.252	0.259	0.226	0.184
KNN Accuracy	0.500	0.500	0.700	0.700	0.772	0.772	0.790	0.790	0.778
KNN Precision	0.264	0.264	0.349	0.349	0.469	0.469	0.649	0.649	0.429
KNN Recall	0.709	0.709	0.413	0.413	0.238	0.238	0.108	0.108	0.014
KNN F1	0.385	0.385	0.378	0.378	0.316	0.316	0.185	0.185	0.026
KNN MCC	0.126	0.126	0.183	0.183	0.212	0.212	0.201	0.201	0.042

(LR: logistic regression, RF: random forest, XGB: XGBoost, KNN: k-nearest neighbors)

F1-Score, and MCC—for each model, as summarized in Table 5. This evaluation provides a dual perspective by leveraging both tabular results and 3D visualizations in Figure 9a to Figure 9e to comprehensively assess model performance under varying threshold settings. The integration of these two approaches highlights the critical role of threshold optimization in tailoring machine learning models to business data mining objectives. From Table 8, it is evident that LR achieved its best Accuracy (0.80) at a threshold of 0.5, whereas RF peaked at a slightly lower value of 0.787 at the same threshold. Interestingly, KNN exhibited the highest accuracy at 0.790 when the threshold was set to 0.7, suggesting its robustness under stricter classification criteria. Precision results also revealed that RF achieved a perfect score (1.00) at a threshold of 0.9, outperforming all other models. However, this high Precision came at the cost of Recall, where LR (0.924 at 0.1 threshold) and KNN (0.709 at 0.1 threshold) performed better, indicating their suitability for tasks requiring exhaustive positive detection. F1-Score analysis reflected a balanced view, with XGB and LR achieving comparable F1 values of 0.419 and 0.428, respectively, at their respective thresholds. MCC, a key metric for imbalanced data, showed XGB (0.287 at 0.5 threshold) and RF (0.307 at 0.4 threshold) leading in terms of overall classification stability. The 3D visualizations from Figure 9a to Figure 9e complement the tabular results by illustrating the interplay between thresholds, models, and metric performance. This analysis underscores the trade-offs associated with dynamic threshold adjustment. While higher thresholds favor metrics like Precision, they may compromise Recall, as seen in RF and KNN's profiles. Conversely, lower thresholds optimize Recall but diminish Precision, as evidenced by LR's profile. Metrics like F1-Score and MCC provide balanced perspectives, capturing these trade-offs effectively.

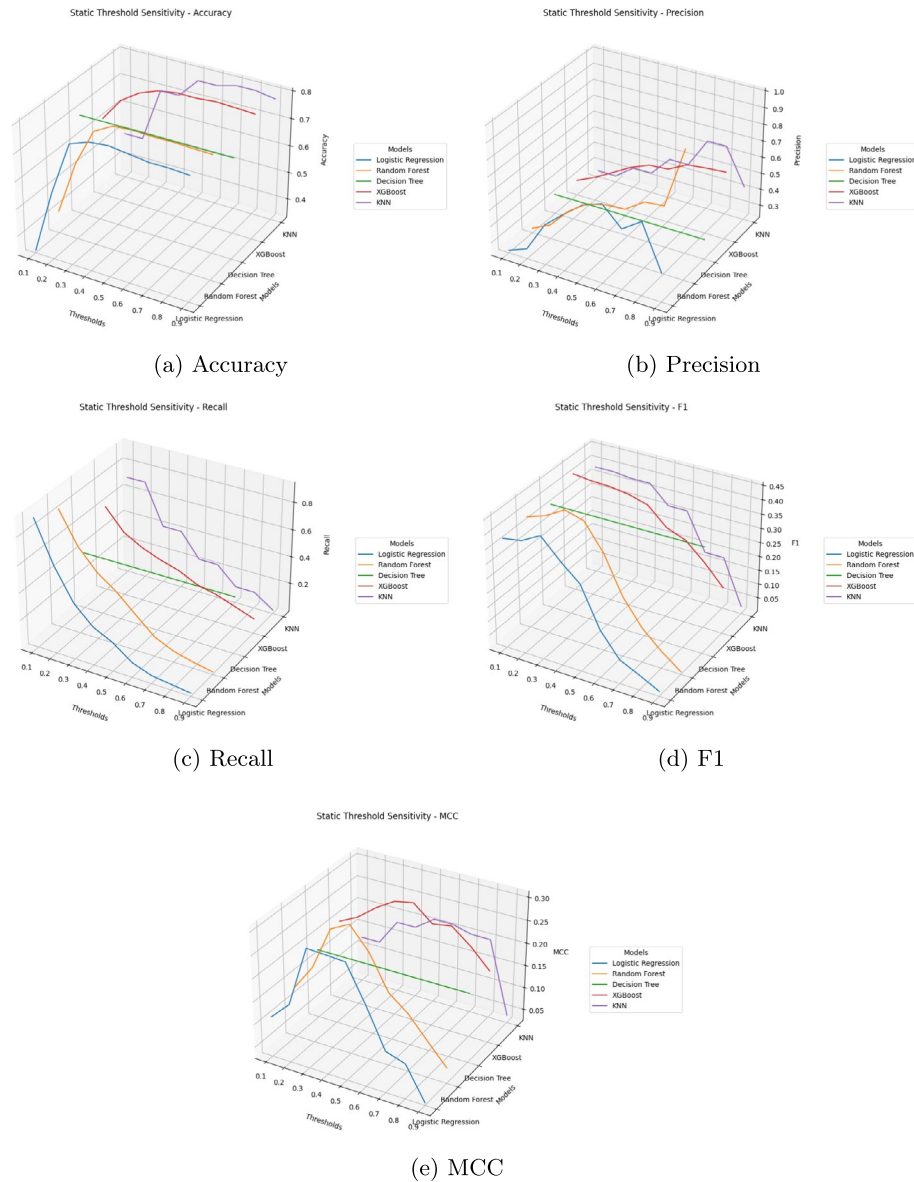


Fig. 8 Static threshold sensitivity analysis across different metrics: **(a)** accuracy, **(b)** precision, **(c)** recall, **(d)** F1, and **(e)** MCC

Before proceeding to robustness and interpretability analysis, we further examine model performance through alternative evaluation metrics and domain-specific cost-sensitive analysis, focusing on the Default of Credit Card Clients dataset.

Alternative metric evaluation

To broaden the depth of our analysis, we evaluated four additional performance metrics that are commonly used in imbalanced classification, such as G-Mean, AUC, Balanced Accuracy, and the IBA with $\alpha = 0.1$. These metrics were calculated for all five classifiers using a fixed decision threshold of 0.5. Table 9 summarizes the results. Among these metrics, DT achieved the highest G-Mean (0.563) and IBA (0.541), indicating relatively balanced performance on both classes. RF performed best on AUC (0.681) and Balanced Accuracy (0.595), while XGB achieved consistent but slightly lower values across all

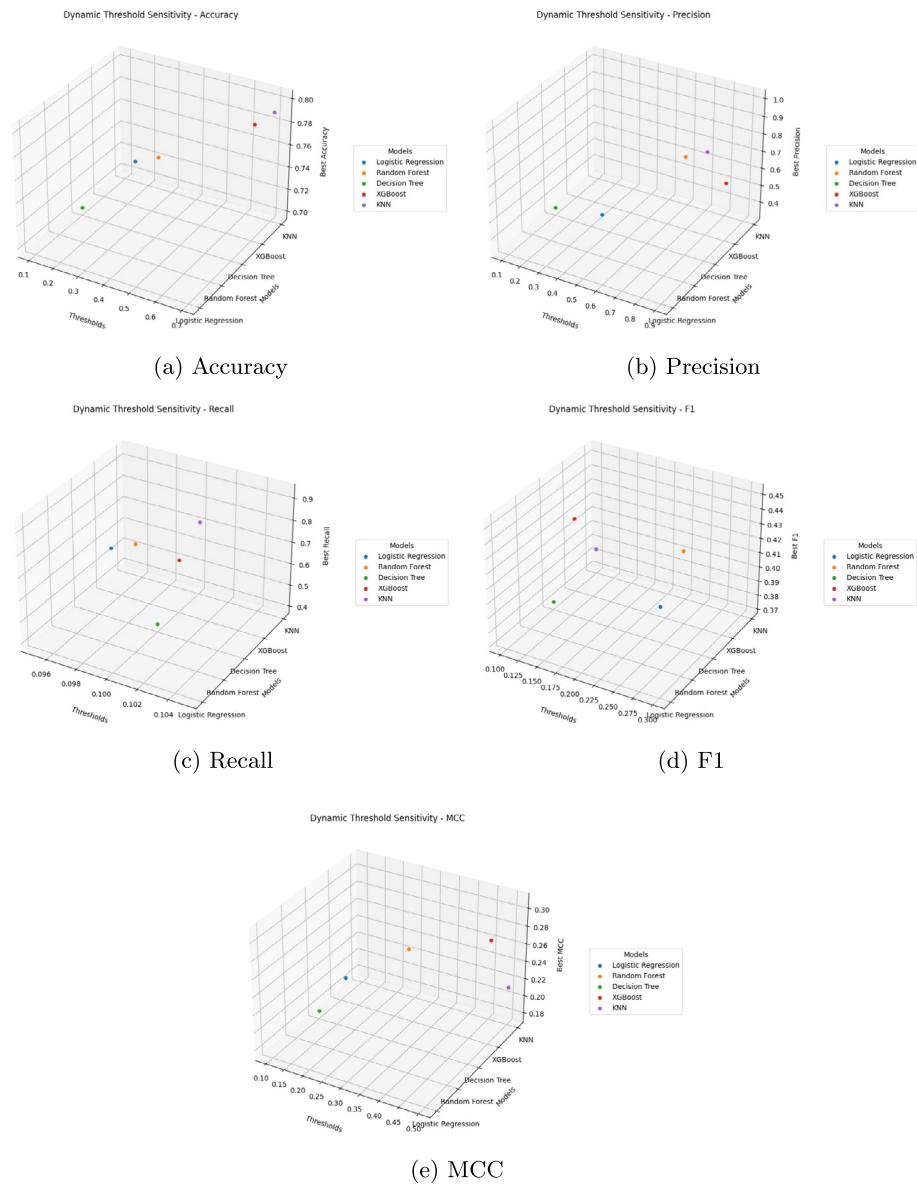


Fig. 9 Dynamic threshold sensitivity analysis comparison across metrics: **(a)** accuracy, **(b)** precision, **(c)** recall, **(d)** F1, and **(e)** MCC

Table 8 Dynamic threshold sensitivity analysis across different metrics

Model	BA	BP	BR	Best F1	Best MCC	BTA	BTP	BTR	BTF	BTM
LR	0.800	0.700	0.924	0.428	0.285	0.5	0.6	0.1	0.3	0.3
RF	0.787	1.000	0.857	0.451	0.307	0.5	0.9	0.1	0.3	0.4
DT	0.700	0.346	0.404	0.373	0.178	0.1	0.1	0.1	0.1	0.1
XGB	0.795	0.636	0.614	0.419	0.287	0.7	0.9	0.1	0.1	0.5
KNN	0.790	0.649	0.709	0.385	0.212	0.7	0.7	0.1	0.1	0.5

Note: LR = Logistic Regression, RF = Random Forest, XGB = XGBoost, KNN = k-Nearest Neighbors, DT = Decision Tree

BA = Best Accuracy, BP = Best Precision, BR = Best Recall

BTA = Best Threshold (Accuracy), BTP = Best Threshold (Precision), BTR = Best Threshold (Recall), BTF = Best Threshold (F1), BTM = Best Threshold (MCC)

metrics. KNN and LR exhibited lower G-Mean and AUC values, reflecting their limited recall performance, as noted earlier. For clarity and interpretability, F1-score and MCC values are also included in Table 9 to facilitate direct comparison. Notably, while some alternative metrics appear competitive, F1-score and MCC remain consistently strong across models, aligning well with overall model behavior. For instance, XGB had the highest F1-score (0.380) and MCC (0.287), along with solid alternative metric values, supporting its well-rounded performance. Given their widespread adoption, statistical interpretability, and alignment with both threshold and cost-based analyses, our selected five metrics were retained as the primary focus for all subsequent robustness, explainability, and statistical validation experiments.

Cost-sensitive evaluation

To assess the alignment between traditional evaluation metrics and real-world financial costs, we performed a cost-sensitive evaluation based on confusion matrix outputs. Each model's false positives and false negatives were penalized using a domain-relevant cost matrix where false positives incurred a cost of 10 units, and false negatives incurred a cost of 100 units. The resulting Expected Cost of Misclassification (ECM) and Net Profit (as the negative of ECM) are reported in Table 10. The results reveal clear differences in cost-effectiveness across models. DT achieved the lowest ECM (15,000) and thus the highest Net Profit (−15,000), primarily due to its higher recall (0.404) and lower false negative count (133). This highlights its suitability for scenarios where missing positive cases (e.g., loan defaults or customer churn) incurs high financial risk. By contrast, LR, which had the highest MCC (0.280) and good precision (0.657), resulted in the highest ECM (18,130) due to a very low recall (0.197) and high number of false negatives (179). This demonstrates that even models with high MCC may not be optimal in cost-sensitive domains if they underperform in recall.

These findings reinforce the decision to prioritize accuracy, precision, recall, F1 and MCC as our core evaluation metrics. While cost-sensitive metrics provide an essential domain-aware perspective, F1-score—by balancing precision and recall—often aligned more closely with cost efficiency, and MCC provided complementary information on class imbalance. Therefore, both metrics remain valid and robust tools for evaluating predictive models in business-critical applications.

Noise robustness testing

The robustness of the models to increasing noise levels (0.0 to 0.3) was evaluated using metrics including Accuracy, Precision, Recall, F1-Score, and MCC. Table 11 details the results, while Figure 10a to Figure 10e illustrate the trends. LR and RF demonstrated stable performance, with LR achieving an accuracy of approximately 0.80, while RF showed a slight improvement to 0.802 at 30% noise. Precision also increased for these models,

Table 9 Performance of standard (F1, MCC) and alternative evaluation metrics across models

Model	F1	MCC	G-Mean	AUC	BA	IBA ($\alpha=0.1$)
RF	0.330	0.253	0.486	0.681	0.595	0.452
XGB	0.380	0.287	0.472	0.665	0.581	0.440
LR	0.303	0.280	0.438	0.653	0.584	0.404
KNN	0.315	0.212	0.469	0.625	0.581	0.436
DT	0.373	0.178	0.563	0.594	0.594	0.541

Table 10 Cost-based evaluation of model performance using confusion matrix analysis

Model	TN	FP	FN	TP	TPR	TNR	FPR	FNR	ECM	Net Profit
LR	764	23	179	44	0.197	0.971	0.029	0.803	18130	−18130
RF	739	48	167	56	0.251	0.939	0.061	0.749	17180	−17180
DT	617	170	133	90	0.404	0.784	0.216	0.596	15000	−15000
XGB	724	63	169	54	0.242	0.920	0.080	0.758	17530	−17530
KNN	727	60	170	53	0.238	0.924	0.076	0.762	17600	−17600

True Positive Rate (TPR), True Negative Rate (TNR), False Positive Rate (FPR), False Negative Rate (FNR), and Expected Cost of Misclassification (ECM) are reported

Table 11 Noise robustness testing across metrics

Noise Level	Model	Accuracy	Precision	Recall	F1-Score	MCC
0	LR	0.800	0.657	0.197	0.303	0.280
0	RF	0.787	0.541	0.238	0.330	0.253
0	DT	0.700	0.346	0.404	0.373	0.178
0	XGB	0.790	0.546	0.291	0.380	0.287
0	KNN	0.772	0.469	0.238	0.315	0.212
0.1	LR	0.797	0.641	0.184	0.286	0.263
0.1	RF	0.794	0.576	0.256	0.354	0.282
0.1	DT	0.756	0.433	0.332	0.376	0.231
0.1	XGB	0.781	0.508	0.291	0.370	0.264
0.1	KNN	0.766	0.445	0.238	0.310	0.198
0.2	LR	0.800	0.672	0.184	0.289	0.276
0.2	RF	0.797	0.582	0.287	0.384	0.304
0.2	DT	0.728	0.363	0.309	0.334	0.165
0.2	XGB	0.777	0.493	0.296	0.370	0.256
0.2	KNN	0.777	0.490	0.229	0.312	0.220
0.3	LR	0.797	0.680	0.152	0.249	0.253
0.3	RF	0.803	0.622	0.274	0.380	0.317
0.3	DT	0.764	0.447	0.283	0.346	0.219
0.3	XGB	0.790	0.566	0.211	0.307	0.249
0.3	KNN	0.771	0.457	0.193	0.271	0.183

with LR and RF reaching 0.680 and 0.622, respectively, highlighting their resilience. Recall trends revealed that the DT had the highest initial recall (0.403) but was sensitive to noise, rebounding slightly at higher noise levels. RF balanced precision and recall effectively, achieving the best F1-Score (0.380) and MCC (0.317) under noisy conditions. LR showed steady MCC values (0.28), while XGB and KNN exhibited declines across metrics, indicating reduced noise tolerance. Overall, RF and LR emerged as the most robust models, demonstrating stability and suitability for noisy environments, making them strong candidates for business applications involving data uncertainty.

Statistical and mathematical validation

To ensure the robustness and reliability of the performance metrics across models, we performed repeated-measures ANOVA, Bootstrap analysis to estimate confidence intervals (CIs) and conducted McNemar's test alongside Cohen's Kappa to statistically evaluate model prediction agreement. This comprehensive approach provides both metric stability assessments and inter-model prediction validation.

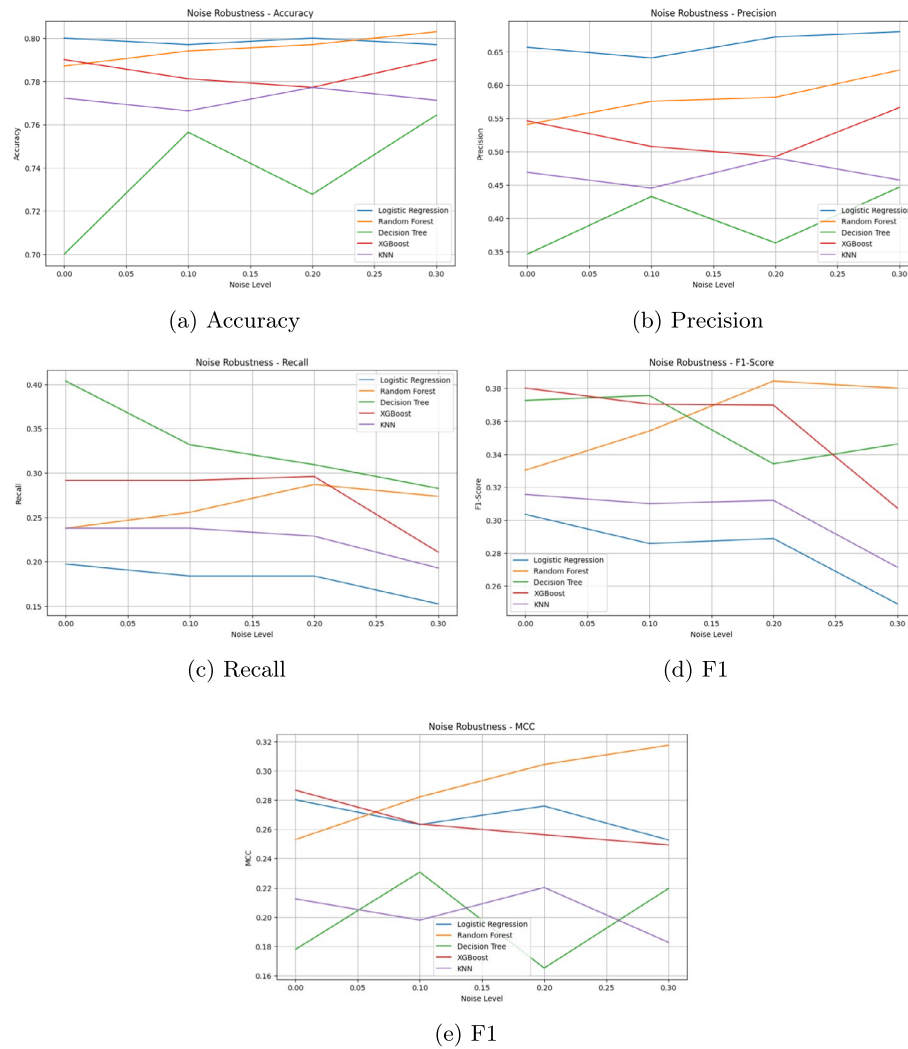


Fig. 10 Noise robustness testing comparison across metrics for all models: **(a)** accuracy, **(b)** precision, **(c)** recall, **(d)** F1, and **(e)** MCC

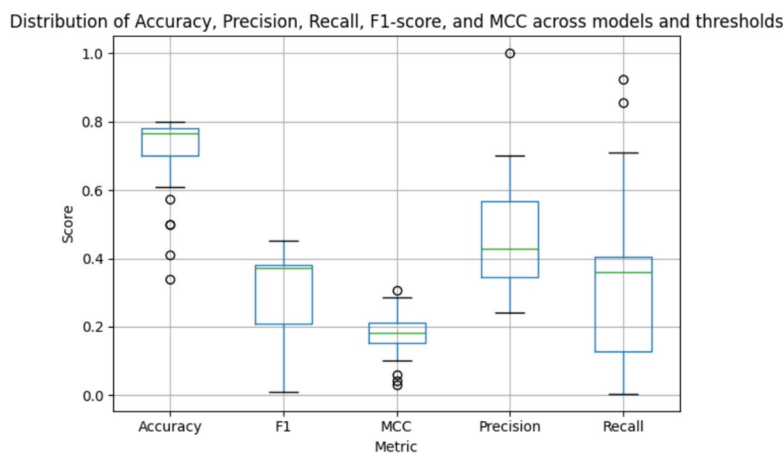
ANOVA

To investigate the stability and reliability of our selected evaluation metrics across models and thresholds, we conducted a repeated-measures ANOVA. The analysis revealed a highly significant main effect of metric on performance scores ($F(4,176) = 77.62$, $p < 0.001$). Descriptive statistics and Holm-corrected post-hoc comparisons are reported in Table 12, and the distribution of scores is visualized in Figure 11. Accuracy achieved the highest overall mean score (0.716) and the lowest variability ($CV = 0.146$), performing significantly better than all other metrics ($p < 0.001$). However, its dominance reflects its bias toward majority classes rather than balanced evaluation under imbalance. F1-score (mean = 0.296, $CV = 0.434$) and MCC (mean = 0.183, $CV = 0.334$) demonstrated more balanced performance across models and thresholds. F1 and Recall did not differ significantly from one another ($p = 0.236$), indicating they capture similar dynamics under imbalance. MCC was significantly lower than F1 and Precision but outperformed Recall, confirming its complementary role alongside F1 in reflecting trade-offs between false positives and false negatives. Precision (mean = 0.462, $CV = 0.340$) performed moderately well, significantly better than Recall and F1 but inferior

Table 12 Descriptive statistics and significance tests for evaluation metrics across models and thresholds

Metric	Mean	Std	Median	IQR	CV	Significant Differences
Accuracy	0.716	0.104	0.767	0.082	0.146	>MCC, Precision, F1, Recall
MCC	0.183	0.061	0.183	0.060	0.334	< Accuracy, F1, Precision; >Recall
Precision	0.462	0.157	0.429	0.221	0.340	< Accuracy; >F1, Recall
F1-score	0.296	0.128	0.373	0.172	0.433	< Accuracy, Precision; >MCC;~Recall (ns)
Recall	0.323	0.228	0.359	0.274	0.708	< Accuracy, Precision;~F1 (ns)

Standard deviation (Std), interquartile range (IQR), and coefficient of variation (CV) are reported. Significant differences are based on Holm-corrected pairwise comparisons at $\alpha = 0.05$

**Fig. 11** Distribution of evaluation metric scores (Accuracy, Precision, Recall, F1-score, and MCC) across five classifiers and nine decision thresholds (0.1–0.9)

to Accuracy. Recall (mean = 0.323, CV = 0.708) was the most unstable metric, with high variability across thresholds and models, making it less reliable as a standalone indicator. Together, these results confirm that although Accuracy yields higher absolute values, F1-score and MCC offer more reliable and interpretable measures under class imbalance, justifying their emphasis in subsequent robustness and explainability analyses in business data mining. Full post-hoc statistics are provided in Table 13.

Bootstrap confidence intervals

Bootstrap CIs for Accuracy, Precision, Recall, F1-Score, and MCC were computed across models. Table 14 presents the Bootstrap confidence interval results across metrics, and Figure 12a to Figure 12e support the visualization of the findings. LR consistently demonstrated narrow intervals across metrics, reflecting its stability. For Accuracy, the CI was (0.7752, 0.8248), higher than DT (0.6723, 0.7287), indicating superior reliability. XGB exhibited tight CIs in most metrics, particularly for MCC (0.2082, 0.3578), suggesting balanced performance. In contrast, DT exhibited wider intervals, especially for Recall, highlighting potential instability under varying data conditions.

McNemar's test and cohen's kappa

McNemar's test evaluated significant prediction differences across paired models. Table 15 and Figure 13 Presents the combined results of McNemar's Test and Cohen's Kappa test. LR and RF showed statistically significant differences in predictions (p-value = 2.4439e-04), indicating non-identical model behaviors. Similarly, DT vs. XGB

Table 13 Pairwise post-hoc comparisons between metrics using repeated-measures ANOVA. Reported values include the t -statistic, raw p -value, and Holm-corrected p -value for family-wise error rate control at $\alpha = 0.05$

Metric A	Metric B	t-stat	p-value	Holm-corrected p	Significant ($\alpha = 0.05$)
Accuracy	MCC	34.220	<0.0001	<0.0001	Yes
Accuracy	Precision	15.056	<0.0001	<0.0001	Yes
Accuracy	F1-score	14.136	<0.0001	<0.0001	Yes
Accuracy	Recall	8.128	<0.0001	<0.0001	Yes
MCC	Precision	-11.227	<0.0001	<0.0001	Yes
MCC	F1-score	7.302	<0.0001	<0.0001	Yes
MCC	Recall	-4.060	0.0002	0.0006	Yes
Precision	F1-score	-4.213	0.0001	0.0005	Yes
Precision	Recall	2.544	0.0145	0.0291	Yes
F1-score	Recall	-1.203	0.2356	0.2356	No

Table 14 Bootstrap confidence intervals for performance metrics across models

Model	Accuracy CI	Precision CI	Recall CI	F1 CI	MCC CI
LR	(0.775, 0.825)	(0.530, 0.768)	(0.148, 0.252)	(0.234, 0.371)	(0.199, 0.355)
RF	(0.762, 0.811)	(0.444, 0.633)	(0.185, 0.293)	(0.263, 0.394)	(0.177, 0.322)
DT	(0.672, 0.729)	(0.287, 0.403)	(0.341, 0.465)	(0.318, 0.424)	(0.110, 0.239)
XGB	(0.763, 0.813)	(0.454, 0.633)	(0.229, 0.348)	(0.312, 0.441)	(0.208, 0.358)
KNN	(0.745, 0.798)	(0.373, 0.570)	(0.181, 0.298)	(0.249, 0.383)	(0.208, 0.358)

(LR: logistic regression, RF: random forest, DT: decision tree, XGB: XGBoost, KNN: k-nearest neighbors)

had a significant p -value of $3.5484\text{e-}22$. Cohen's Kappa values, which quantify agreement, were highest for LR vs. RF (0.5460), suggesting moderate agreement, while RF vs. DT scored lower (0.3172), reflecting greater disparity in predictions.

The bootstrap analysis identified LR and XGB as the most stable models across metrics, with consistently narrow CIs. McNemar's test further validated significant predictive differences between DT and XGB, while Cohen's Kappa indicated strong agreement between LR and RF. These findings underscore LR's reliability for precision-critical tasks and XGB's balance across metrics. DT's variability suggests its use may be better suited for scenarios prioritizing Recall over precision or stability.

Explainable AI (XAI) analysis

To strengthen the interpretability of our framework, we implemented a two-stage SHAP analysis. First, we applied conventional SHAP techniques (bar plots and beeswarm visualizations) to LR as a baseline model, providing standard global and local feature importance insights. This step ensures transparency and aligns with widely adopted explainability practices. Building on this, we introduced a novel 3D metric-conditioned SHAP analysis, where feature contributions were examined across varying thresholds and linked directly to performance metrics (Accuracy, Precision, Recall, F1-score, and MCC). This extension moves beyond static feature importance, offering a dynamic view of how interpretability interacts with metric reliability under class imbalance.

Conventional SHAP analysis

To enhance transparency and meet standard explainability requirements, we first applied conventional SHAP analysis to LR, one of the most interpretable models in our study and a natural baseline for business applications. SHAP values were computed using the

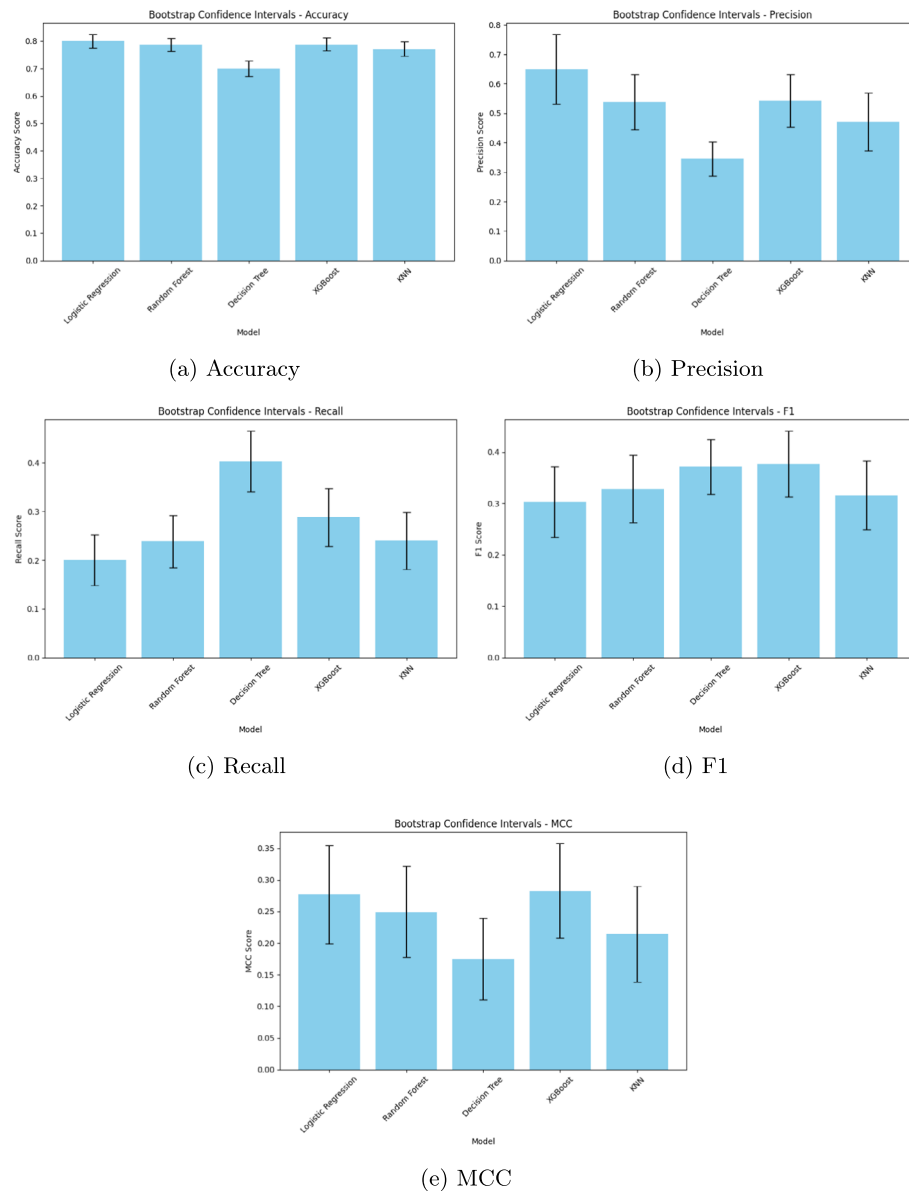


Fig. 12 Bootstrap confidence interval comparison for each metric: (a) accuracy, (b) precision, (c) recall, (d) F1, and (e) MCC

Table 15 Combined results of McNemar’s test and Cohen’s kappa for model comparisons

Model comparison	McNemar p-value	Cohen kappa
LR vs RF	0.00024439	0.546
RD vs DT	3.0985E-32	0.317
DT vs XGB	3.5484E-22	0.291
XGB vs KNN	0.63977	0.444

LinearExplainer, with results visualized as global feature importance rankings in Figure 14a and beeswarm distributions of individual SHAP contributions in Figure 14b. The results confirm that repayment history (PAY_0, PAY_3, PAY_5) and billing amounts (BILL_AMT1–3, BILL_AMT5) are the most influential predictors of customer default, consistent with financial domain expectations. In contrast, demographic variables such

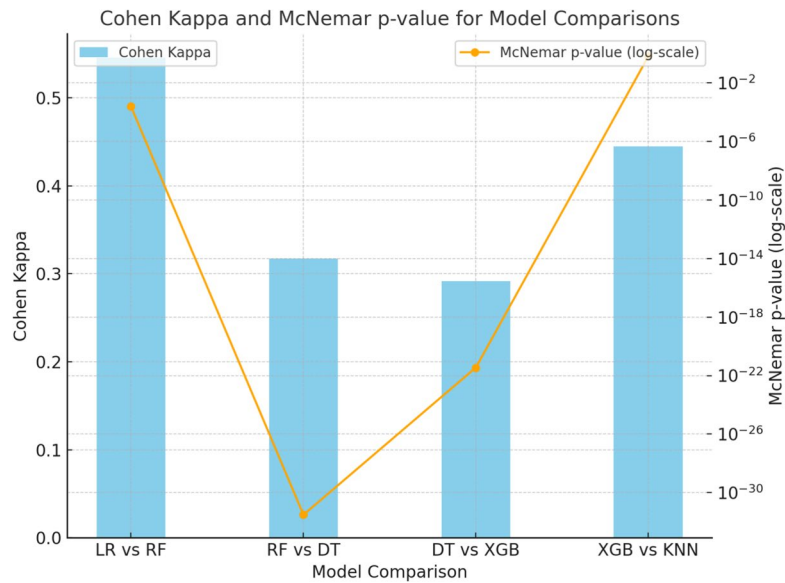


Fig. 13 McNemar's test and Cohen's kappa comparisons

as SEX and EDUCATION exerted negligible influence, highlighting the model's reliance on behavioral rather than demographic signals. Importantly, these SHAP explanations provide context for the evaluation metrics examined in this study. For example, the strong dominance of repayment-related variables explains why Recall remains unstable under threshold changes (since minority-class detection depends heavily on a few repayment indicators). Similarly, the broader balance of billing and repayment features supports the complementary nature of F1-score and MCC, which integrate precision and sensitivity trade-offs.

By combining conventional SHAP outputs with metric-specific analyses, our framework demonstrates how interpretability tools can bridge the gap between raw performance metrics and domain-level decision requirements, ensuring that metric selection is not only statistically rigorous but also practically explainable.

Metric-conditioned SHAP analysis (3D threshold interaction)

To move beyond static feature importance, we applied our novel 3D SHAP analysis framework to investigate how individual features interact with evaluation metrics across varying thresholds. The 3D visualizations in Figure 15a through Figure 15e show SHAP importance (Y-axis) plotted against decision thresholds (X-axis) and metric scores (Z-axis), enabling a dynamic interpretability view tailored to each performance metric.

In Figure 15a, which represents Accuracy, features like LIMIT_BAL and PAY_0 consistently demonstrate high SHAP importance across threshold values. This stability suggests these features reliably contribute to correct classifications regardless of decision boundaries, making accuracy-sensitive models suitable for high-confidence applications such as credit approval or transaction filtering. Precision in Figure 15b reveals a more selective pattern. SHAP contributions from LIMIT_BAL and PAY_0 peak at mid-range thresholds (0.4–0.6), aligning with the metric's preference for minimizing false positives. However, contributions from other features such as BILL_AMT1 and PAY_3 sharply decline as thresholds increase, highlighting the trade-off precision makes at higher thresholds, often at the cost of sensitivity. The Recall visualization in Figure 15c presents

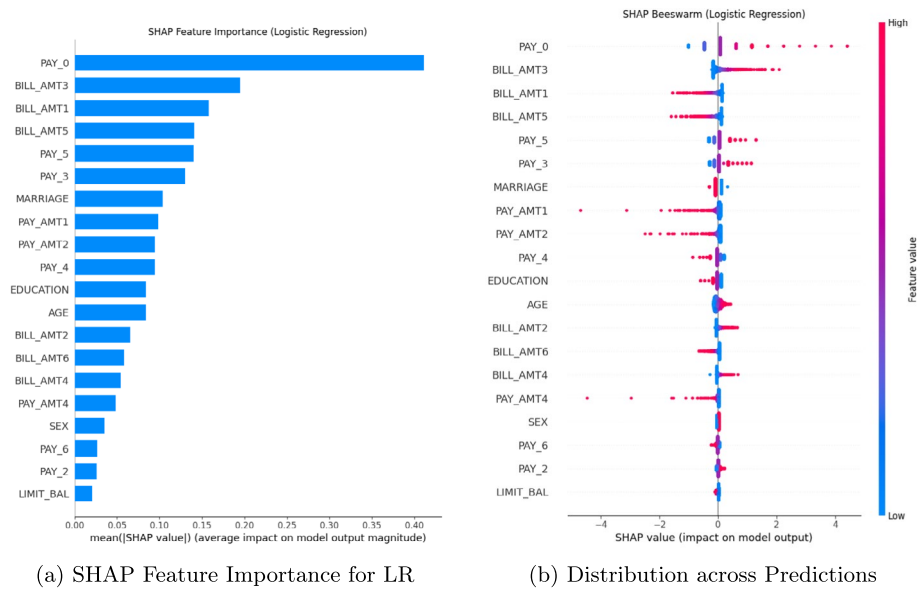


Fig. 14 SHAP-based model interpretability for Logistic Regression. **(a)** Global feature importance based on mean absolute SHAP values. **(b)** Beeswarm plot showing the distribution of SHAP values across individual predictions

a contrasting behavior. Here, SHAP values for almost all features, particularly PAY_0, AGE, and PAY_3, decrease rapidly as thresholds rise. This is consistent with recall’s sensitivity to false negatives; as the model becomes more conservative, it misses more positive cases, and the features driving true positive capture lose influence. For business use cases focused on early risk detection or churn prevention, such patterns underscore the cost of threshold tuning on recall-driven models.

In Figure 15d, the F1-score—being the harmonic mean of precision and recall—exhibits a more balanced SHAP profile. Features like PAY_0, BILL_AMT2, and PAY_AMT1 maintain moderately high importance across thresholds, particularly in the 0.3–0.6 range where the F1-score tends to peak. This balanced feature attribution reinforces F1’s utility in capturing both class sensitivity and specificity, making it a stable and adaptable metric for business contexts with imbalanced classes.

The MCC plot in Figure 15e further validates these observations. LIMIT_BAL, PAY_0, and PAY_AMT1 show consistently strong SHAP contributions across thresholds, indicating that MCC—like F1—captures balanced class performance but with additional sensitivity to all four confusion matrix categories. Notably, MCC also reveals subtler contributions from secondary features like AGE and PAY_3, which are less prominent under metrics like Precision or Accuracy.

Taken together, these visualizations illustrate how each metric aligns with different feature behaviors under threshold shifts. Our 3D SHAP analysis not only enhances model interpretability but also provides practical guidance: F1-score and MCC emerge as the most stable and balanced metrics for business classification under class imbalance, while Accuracy and Precision may be preferable when robustness or specificity are prioritized. This analysis empowers practitioners to align metric choice with both performance goals and feature behavior.

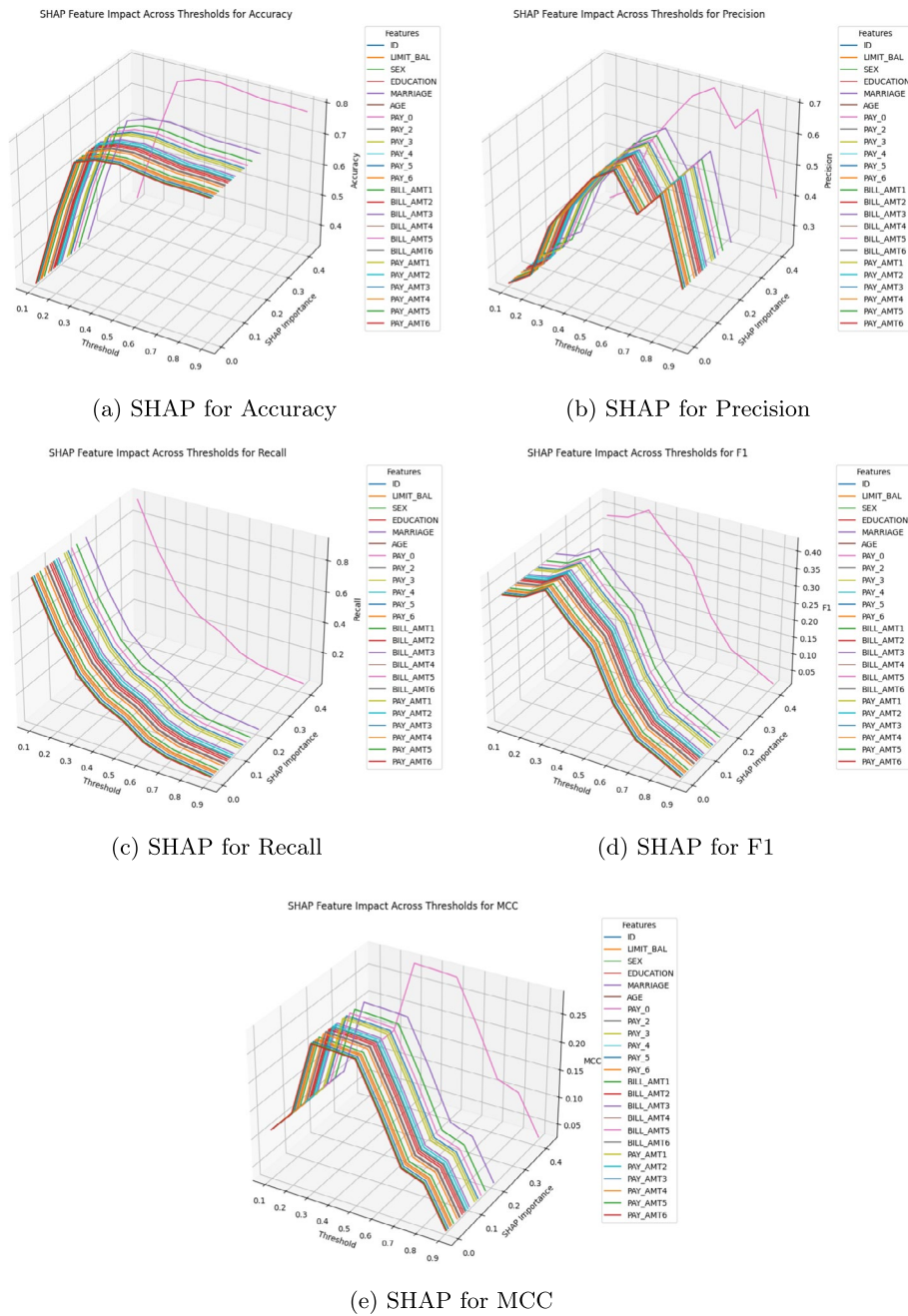


Fig. 15 SHAP feature impact analysis across thresholds: (a) SHAP for accuracy, (b) SHAP for precision, (c) SHAP for recall, (d) SHAP for F1, and (e) SHAP for MCC

Additional experiment on telco customer churn dataset

To further validate the robustness and generalizability of our results, we extended our analysis to the Telco Customer Churn dataset (7,043 instances and 20+ features), sourced from Kaggle. This dataset addresses a binary classification problem predicting customer churn, where the target variable denotes whether a customer has left ('Yes') or remained ('No'). The dataset is imbalanced, with approximately 26.5% of customers having churned, making it suitable for examining the performance of models on imbalanced data.

Cross-validation results

The cross-validation results in Table 16 highlight that LR achieved the highest Accuracy (0.803) but had a relatively low Recall (0.548), indicating its inability to effectively predict churned customers. RF performed slightly lower with Accuracy (0.793) but showed similarly low Recall (0.493), which is not ideal for imbalanced classification problems. In contrast, XGB achieved the highest F1-score (0.562) by maintaining a good balance between Precision (0.611) and Recall (0.521). This suggests that the F1-score, which balances both Precision and Recall, is a highly effective metric in business data mining for imbalanced datasets. KNN and DT performed poorly in comparison, especially in Precision and MCC, underlining their limited suitability for this task.

Noise robustness testing

The noise robustness testing results Table 17 indicate that XGB remained stable even with increasing noise levels (0.0 to 0.3), maintaining solid F1-scores and MCC values. This resilience to noise makes XGB a robust model for real-world business applications, where noisy data is common. LR and RF showed relatively stable MCC values under noise conditions, further supporting their reliability. However, KNN and DT showed more sensitivity to noise, with a more significant decline in performance.

Bootstrap confidence intervals

The bootstrap confidence intervals for each model Table 18 further validate XGB's reliability. The F1-score and MCC for XGB showed the most stable confidence intervals, with F1-score CI ranging from 0.2141 to 0.2968 and MCC CI from 0.2279 to 0.3550. In comparison, LR and RF showed wider intervals, particularly for Recall and MCC, indicating more variability in performance. The narrow confidence intervals for XGB reinforce the conclusion that F1-score and MCC are the best metrics for evaluating imbalanced datasets in business data mining.

McNemar's test & cohen's kappa

Statistical significance between model pairs was assessed using McNemar's test and Cohen's Kappa Table 19. XGB showed significant performance differences compared to other models, particularly in its Precision and Recall, as evidenced by the McNemar p-value of 0.0245 in the comparison with KNN. Cohen's Kappa confirmed that XGB had higher agreement with other models, particularly in predicting churn, further highlighting its superiority in this task.

Metric-conditioned SHAP analysis with telco customer churn dataset

To understand the influence of feature importance on model decisions across different performance metrics, we conducted SHAP-based explainable AI analysis using 3D

Table 16 Cross-validation results for the Telco Customer Churn dataset, including accuracy, precision, recall, F1-score, and MCC

Model	Accuracy	Precision	Recall	F1	MCC
LR	0.803	0.654	0.548	0.596	0.470
RF	0.793	0.645	0.493	0.558	0.430
DT	0.734	0.498	0.520	0.510	0.330
XGB	0.785	0.611	0.521	0.562	0.420
KNN	0.761	0.553	0.522	0.537	0.380

Table 17 Noise robustness testing across metrics for the Telco Customer Churn dataset

Noise Level	Model	Accuracy	Precision	Recall	F1-Score	MCC
0.0	LR	0.804	0.648	0.575	0.609	0.480
0.0	RF	0.788	0.620	0.527	0.569	0.433
0.0	DT	0.719	0.472	0.468	0.470	0.279
0.0	XGB	0.767	0.568	0.516	0.541	0.386
0.0	KNN	0.756	0.540	0.540	0.540	0.374
0.1	LR	0.805	0.653	0.570	0.609	0.482
0.1	RF	0.792	0.636	0.505	0.563	0.434
0.1	DT	0.757	0.544	0.524	0.534	0.370
0.1	XGB	0.774	0.568	0.628	0.596	0.441
0.1	KNN	0.753	0.535	0.527	0.531	0.363
0.2	LR	0.802	0.650	0.556	0.599	0.472
0.2	RF	0.798	0.651	0.519	0.577	0.452
0.2	DT	0.731	0.496	0.588	0.538	0.353
0.2	XGB	0.791	0.640	0.489	0.555	0.428
0.2	KNN	0.748	0.525	0.556	0.540	0.368
0.3	LR	0.798	0.641	0.548	0.591	0.460
0.3	RF	0.792	0.646	0.479	0.550	0.426
0.3	DT	0.767	0.564	0.546	0.554	0.397
0.3	XGB	0.766	0.555	0.610	0.581	0.420
0.3	KNN	0.758	0.544	0.559	0.552	0.386

Table 18 Bootstrap confidence intervals with the Telco Customer Churn dataset for performance metrics across models

Model	Accuracy CI	Precision CI	Recall CI	F1 CI	MCC CI
LR	(0.601, 0.645)	(0.220, 0.307)	(0.199, 0.275)	(0.209, 0.284)	(−0.051, 0.050)
RF	(0.607, 0.654)	(0.221, 0.318)	(0.187, 0.257)	(0.199, 0.281)	(0.199, 0.281)
DT	(0.589, 0.632)	(0.221, 0.277)	(0.224, 0.303)	(0.228, 0.304)	(0.228, 0.304)
XGB	(0.597, 0.645)	(0.216, 0.312)	(0.205, 0.272)	(0.214, 0.297)	(0.228, 0.355)
KNN	(0.586, 0.633)	(0.213, 0.309)	(0.224, 0.305)	(0.228, 0.355)	(0.228, 0.355)

(LR: logistic regression, RF: random forest, DT: decision tree, XGB: XGBoost, KNN: k-nearest neighbors)

Table 19 Results of McNemar’s test and Cohen’s kappa for pairwise model comparisons on the Telco Customer Churn dataset

Model Comparison	McNemar p-value	Cohen’s Kappa
LR vs RF	0.101	0.669
RF vs DT	0.000	0.527
DT vs XGB	0.058	0.528
XGB vs KNN	0.025	0.595

visualizations. Figure 16a through Figure 16e illustrate the variation of SHAP values with respect to classification thresholds and their respective impact on Accuracy, Precision, Recall, F1-score, and MCC. Each figure represents a 3D plot where the x-axis denotes the classification threshold, the y-axis corresponds to the SHAP value (indicating feature importance), and the z-axis presents the value of the respective performance metric. In Figure 16a, representing SHAP impact on Accuracy, the curves reveal that Accuracy peaks around threshold values of 0.5 to 0.6, but varies significantly with feature influence. Notably, features such as InternetService_No, MonthlyCharges, and tenure dominate the model’s predictive power. However, the sharp drops in accuracy beyond certain thresholds suggest sensitivity to threshold tuning and class imbalance, indicating that Accuracy alone may be unreliable in imbalanced business scenarios. Figure 16b

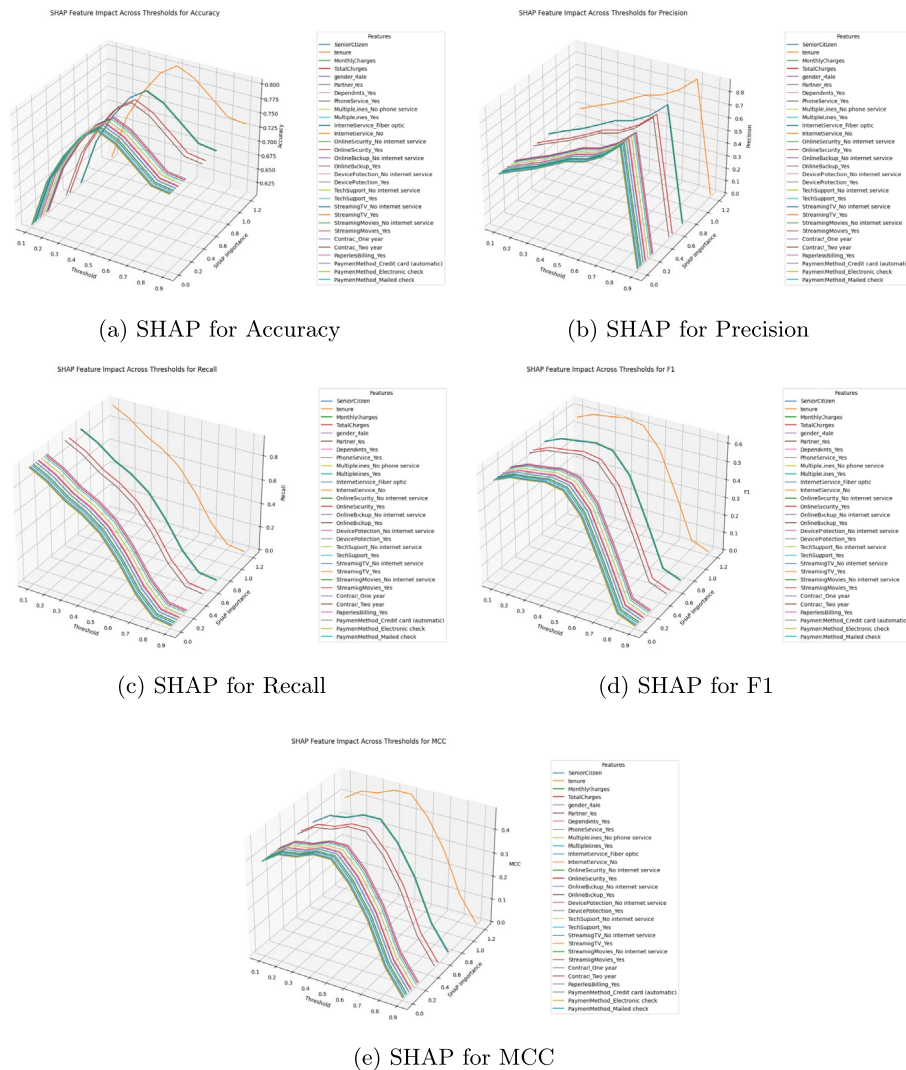


Fig. 16 SHAP feature impact analysis across thresholds with the Telco Customer Churn dataset: **(a)** SHAP for accuracy, **(b)** SHAP for precision, **(c)** SHAP for recall, **(d)** SHAP for F1, and **(e)** SHAP for MCC

demonstrates SHAP impact on Precision. Precision shows a noticeable spike at higher thresholds (around 0.7), but its distribution across features is more uneven. Certain features, especially InternetService_No and ElectronicCheck, produce sharp increases in SHAP values, indicating high but potentially unstable influence. This suggests that while Precision may seem high at certain thresholds, it may favor false negatives and be misleading without recall context. In Figure 16c, we observe the SHAP impact on Recall. Here, most features demonstrate a steady decline in recall as the threshold increases, confirming that lower thresholds tend to favor true positive recovery. While this stability is beneficial, high recall alone without balancing false positives may not reflect overall predictive reliability, particularly for business-critical applications like churn detection. Figure 16d provides insights into F1-score, which balances Precision and Recall. The SHAP curves here are more evenly distributed, and the metric reaches optimal values at thresholds around 0.4 to 0.6. This balanced behavior across features and thresholds suggests that F1-score is more robust and interpretable than single-dimension metrics, making it highly suitable for imbalanced business data mining problems. Finally,

Figure 16e illustrates the SHAP analysis for MCC. Similar to F1-score, MCC also displays smooth curves across thresholds and SHAP values, with consistent contributions from core features such as tenure, MonthlyCharges, and Contract_Two year. The metric's stability across threshold changes further confirms its value as a balanced and reliable indicator of classification performance under imbalanced conditions. Overall, these SHAP-based visualizations reinforce our conclusion that both F1-score and MCC provide the most stable and interpretable metrics for model evaluation in business data mining. Unlike Accuracy and Precision, which can mislead in the presence of skewed class distributions, F1 and MCC reflect more nuanced, threshold-robust insights into model behavior and feature influence.

The additional experiments on the Telco Customer Churn dataset reinforce our earlier findings and provide further evidence for effective model and metric selection in imbalanced business classification. Among the evaluated models, XGB consistently outperformed others across cross-validation, noise robustness, bootstrap confidence intervals, and statistical testing, making it the most reliable choice for churn prediction. F1-score and MCC again proved superior to traditional metrics like accuracy and precision, showing strong stability across thresholds and resilience to data noise. SHAP-based explainability confirmed these results, with 3D SHAP visualizations revealing that F1 and MCC captured consistent, interpretable feature contributions—particularly from tenure, MonthlyCharges, and contract type—regardless of threshold shifts. In contrast, accuracy and precision were more sensitive to individual features and less stable across decision boundaries. Overall, this extended evaluation confirms that XGB, combined with F1-score and MCC, provides a robust and interpretable solution for business-critical tasks like churn prediction. These results highlight the importance of using advanced metrics and explainability tools when developing decision-support systems in imbalanced, high-stakes domains.

Discussion

This study presents a comprehensive evaluation of performance metric reliability for imbalanced classification in business data mining. By analyzing two real-world datasets of differing sizes and domains—the large-scale Default of Credit Card Clients dataset (30,000 instances) and the moderately sized Telco Customer Churn dataset (7,043 instances)—we systematically assessed five machine learning models (LR, RF, DT, XGB, and KNN) across a variety of metrics, threshold strategies, noise conditions, and interpretability techniques.

Our findings reinforce the importance of aligning metric selection with both the intrinsic data characteristics and the operational goals of business decision-making. Across both datasets, we observed that threshold selection plays a pivotal role in determining metric behavior. Static threshold analysis revealed that Accuracy and Precision typically peaked at moderate thresholds (0.5–0.7), but consistently failed to account for minority class detection, favoring specificity over sensitivity. Conversely, Recall peaked at lower thresholds (e.g., 0.1) but rapidly degraded with threshold increases, highlighting its trade-off bias and limited practical utility when used in isolation. These findings were further validated through dynamic threshold sensitivity analysis, which allowed us to identify the exact thresholds at which each metric achieved optimal performance. Notably, both F1-score and MCC achieved their best balance between false positives and

false negatives at thresholds between 0.3 and 0.5, a result consistently observed across both datasets and model architectures [63].

Robustness testing under varying levels of synthetic noise offered further insights into metric stability. As expected, LR and RF exhibited relatively stable Accuracy and Precision across noise conditions, particularly in the Telco dataset, where the signal-to-noise ratio was narrower. DT performance, however, fluctuated considerably. Importantly, Recall emerged as the most noise-sensitive metric, often producing unstable outputs in both datasets. In contrast, F1-score and MCC maintained more consistent performance across all noise levels, reaffirming their value for real-world applications where data imperfections are common [64].

To establish statistical validity, we conducted ANOVA and McNemar's tests across all model-metric combinations [65]. Accuracy and Precision often failed to yield statistically significant differences between models, underscoring their limited discriminatory power in imbalanced settings [66]. On the other hand, F1-score and MCC showed significant inter-model variation across both datasets. For instance, the F1-score was statistically superior when comparing RF to LR on the credit default dataset, while MCC significantly differentiated RF and XGB on the churn dataset [67]. These results demonstrate that F1 and MCC are not only more stable but also more diagnostically useful for comparative model evaluation in imbalanced classification.

Beyond performance scores, we incorporated a two-stage explainable AI (XAI) analysis using SHAP to examine feature contributions across evaluation metrics and decision thresholds. In the first stage, conventional SHAP plots (e.g., bar and beeswarm) applied to the LR model identified key predictive drivers—such as PAY_0 and LIMIT_BAL in the credit default dataset, and tenure and Contract type in the churn dataset—consistent with domain expectations [68]. In the second stage, we introduced a novel 3D SHAP framework to assess how feature importance evolved across thresholds and metrics. This dynamic analysis revealed that Accuracy and Precision were driven by high-specificity features whose contributions declined at extreme thresholds, while F1-score and MCC exhibited more stable and distributed feature influence. This dual-layer XAI approach enhanced both global and threshold-specific interpretability, reinforcing the practical advantage of metrics like F1 and MCC in real-world decision contexts where model transparency is essential [69].

From a business perspective, the cost of false negatives—such as missed loan defaults or undetected customer churn—is often significantly higher than that of false positives. In such high-stakes contexts, the F1-score offers a reliable trade-off metric by balancing precision and recall. MCC further strengthens evaluation by considering all elements of the confusion matrix, making it especially useful in operational settings where misclassification costs are asymmetrical but not always explicitly defined.

To broaden our analysis, we also evaluated four alternative metrics commonly used in imbalanced classification: AUC, G-Mean, Balanced Accuracy, and the Index of Balanced Accuracy (IBA, with $\alpha = 0.1$). Models such as DT and RF performed well under these metrics, particularly in terms of G-Mean and AUC. However, these alternatives lacked the statistical consistency and interpretive robustness observed with F1-score and MCC, reinforcing our choice to prioritize the latter in our main analysis.

In addition, we implemented a cost-sensitive evaluation framework that applied business-relevant penalties to misclassifications using Expected Cost of Misclassification

(ECM) and Net Profit. This revealed that models with higher recall—such as the DT—achieved lower overall cost, despite having lower MCC or precision scores. These findings underscore the importance of aligning metric selection not only with data characteristics but also with domain-specific cost structures. Together, they reaffirm F1-score and MCC as both statistically sound and operationally meaningful metrics for imbalanced business classification.

In conclusion, our study proposes a statistically grounded and interpretable framework for selecting performance metrics in business data mining. By integrating dynamic thresholding, noise robustness testing, inferential statistics, and SHAP-based explainability, we provide a generalizable methodology for model evaluation on imbalanced datasets. This approach equips practitioners and researchers with the tools needed to make evidence-based decisions in predictive modeling, ensuring both robustness and transparency in high-stakes applications. Future work can extend this framework to unstructured data types or incorporate real-time cost-aware decision systems in dynamic business environments.

Limitations

While our study offers a comprehensive and statistically rigorous framework for evaluating performance metrics in imbalanced business data mining, several limitations should be acknowledged. First, although we examined two real-world business datasets—one large-scale (credit card default) and one smaller-scale (customer churn)—both originate from structured tabular domains. Future research should consider a broader range of dataset sizes, structures, and industries to better assess the generalizability of our findings. Second, our evaluation focused on five widely used machine learning models, including both linear and tree-based methods. However, we did not incorporate deep learning architectures, which may yield different patterns in both performance and explainability—particularly in high-dimensional or unstructured data contexts. Third, while our primary analysis concentrated on five core evaluation metrics—Accuracy, Precision, Recall, F1-score, and MCC—we also included supplementary results for alternative metrics such as AUC, G-Mean, Balanced Accuracy, and IBA. These additional metrics were analyzed descriptively to broaden the scope of our findings but were not subjected to the same level of statistical testing or interpretability analysis. Future studies could more deeply investigate the statistical robustness and explainability of these alternative metrics. Fourth, although we introduced a cost-sensitive evaluation framework and analyzed Expected Cost of Misclassification (ECM) and Net Profit, we did not incorporate other domain-specific constraints such as real-time decision thresholds, risk-based optimization, or adaptive cost matrices. These remain important areas for operationalizing evaluation metrics in live business systems. Finally, this study focused on classification-based business predictive modeling. Extending the framework to other decision-support contexts such as forecasting, optimization, or recommendation remains an open avenue for future work.

Conclusion

This comprehensive research presents a statistically grounded and interpretable framework for selecting performance metrics in business data mining tasks involving imbalanced datasets. Using two real-world benchmarks—the Default of Credit Card Clients

and the Telco Customer Churn datasets—we evaluated five machine learning models through a multidimensional approach incorporating cross-validation, threshold sensitivity, noise robustness, inferential statistics, and SHAP-based explainability. Our results show that Accuracy and Precision are inadequate in imbalanced contexts, as they favor the majority class and underrepresent minority outcomes. Recall, though more sensitive to the minority class, showed instability across thresholds and noise levels. In contrast, F1-score demonstrated the most reliable and balanced performance across all scenarios, while MCC offered complementary insights by incorporating all elements of the confusion matrix. The addition of a two-layer SHAP framework linked feature importance to metric behavior, highlighting how F1-score and MCC maintain interpretability and feature stability across thresholds. This supports their use in business-critical applications where transparency and robustness are essential. Future research should extend this framework to multi-class problems, cost-sensitive learning, and deep learning models tailored for skewed data. Incorporating real-world business constraints such as monetary costs, regulatory compliance, or real-time adaptation will further enhance practical utility. In summary, this work provides a comprehensive, interpretable, and transferable methodology for metric evaluation in imbalanced classification—empowering practitioners to make evidence-based, risk-aware decisions in predictive modeling.

Author contributions

KMS perceived the idea, carried out formal analysis, developed the visualization and methodology, and wrote the draft. RH investigated, validated, provided resources, and supervised the study. KC and MAS administrated the project, and participated in reviewing the initial draft, modified it, and considerably improved it.

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Data availability

The datasets analyzed during this study are publicly available at <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients> (accessed on 16 Dec. 2024), and at <https://www.kaggle.com/datasets/blastchar/telco-customer-churn> (accessed on 11 Jun. 2025).

Declarations

Competing interests

The authors declare no competing interests.

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References

1. Bhatia S, Sharma P, Burman R, Hazari S, Hande R. Credit scoring using machine learning techniques. *Int J Comput Appl*. 2017;161(11):1–4.
2. Saito T, Rehmsmeier M. The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*. 2015. <https://doi.org/10.1371/journal.pone.0118432>.
3. Juba B, Le HS. Precision-recall versus accuracy and the role of large data sets. *Proceedings of the AAAI Conference on Artificial Intelligence*. 2019;33:4039–48.
4. Boughorbel S, Jarray F, El-Anbari M. Optimal classifier for imbalanced data using matthews correlation coefficient metric. *PLoS ONE*. 2017. <https://doi.org/10.1371/journal.pone.0177678>.
5. Xu X, Chen W, Sun Y. Over-sampling algorithm for imbalanced data classification. *JSEE*. 2019. <https://doi.org/10.21629/jsee.2019.06.12>.
6. Shaer L, Kanj R, Joshi R. Data imbalance handling approaches for accurate statistical modeling and yield analysis of memory designs. 2019 IEEE International Symposium on Circuits and Systems (ISCAS), 1–5 2019 <https://doi.org/10.1109/IS-CAS.2019.8702731>
7. Wu X, Huang F, Huang H. Fast stochastic recursive momentum methods for imbalanced data mining. 2022 IEEE International Conference on Data Mining (ICDM), 578–587 (2022) <https://doi.org/10.1109/ICDM54844.2022.00068>

8. Sujon KM, Hassan R, Khairudin AR, Moi SH, Shafie MLM, Saringat Z, et al. The effects of imbalanced datasets on machine learning algorithms in predicting student performance. *JOIV: International Journal on Informatics Visualization*. 2024;8(3–2):1599–605.
9. Japkowicz N. Assessment metrics for imbalanced learning. *Imbalanced learning: Foundations, algorithms, and applications*, 2013;187–206.
10. Ferri C, Hernández-Orallo J, Modrou R. An experimental comparison of performance measures for classification. *Pattern Recogn Lett*. 2009;30(1):27–38.
11. Kubat M. Addressing the curse of imbalanced training sets: one-sided selection. In: *Proceedings of the 14th International Conference on Machine Learning*, pp. 179–186 (1997). Morgan Kaufmann.
12. Diallo R, Edalo C, Awe OO. Machine learning evaluation of imbalanced health data: A comparative analysis of balanced accuracy, mcc, and f1 score. In: Awe, O.O., Vance, E.A. (eds.) *Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network*, USA. STEAM-H: Science, Technology, Engineering, Agriculture, Mathematics & Health, pp. 283–312. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-72215-8_12
13. Zareapoor M, Shamsolmoali P. Boosting prediction performance on imbalanced dataset. *Int J Inf Commun Technol*. 2018;13:186–95. <https://doi.org/10.1504/IJICT.2018.10011701>.
14. Imran M, Qyser A, Ali SS, Kumar V, Jah M, Malla N. An overview on data mining designed for imbalanced datasets. *International Journal of Research in Engineering and Technology*. 2014;03:222–5. <https://doi.org/10.15623/IJRET.2014.0310034>.
15. Chakraborty T. Imbalanced ensemble classifier for learning from imbalanced business school dataset. *International Journal of Mathematical, Engineering and Management Sciences* 2018; <https://doi.org/10.33889/IJMEMS.2019.4.4-068>
16. Lee Z, Lee C-Y, Chou S-T, Ma W-P, Ye F, Chen Z. A hybrid system for imbalanced data mining. *Microsyst Technol*. 2020;26:3043–7. <https://doi.org/10.1007/S00542-019-04566-1>.
17. Syaripudin A, Khodra ML. A comparison for handling imbalanced datasets. 2014 International Conference of Advanced Informatics: Concept, Theory and Application (ICAICTA), 293–298 (2014) <https://doi.org/10.1109/ICAICTA.2014.7005957>
18. Vluymans S. Learning from imbalanced data. *Dealing with Imbalanced and Weakly Labelled Data in Machine Learning using Fuzzy and Rough Set Methods*. 2018. https://doi.org/10.1007/978-3-030-04663-7_4.
19. Yan Y, Liu Y, Shyu M, Chen M. Utilizing concept correlations for effective imbalanced data classification. *Proceedings of the 2014 IEEE 15th International Conference on Information Reuse and Integration (IEEE IRI 2014)*, 561–568 (2014) <https://doi.org/10.1109/IRI.2014.7051939>
20. Veganzones D, Séverin E. An investigation of bankruptcy prediction in imbalanced datasets. *Decis Support Syst*. 2018;112:111–24. <https://doi.org/10.1016/j.dss.2018.06.011>.
21. Susan S, Kumar A. The balancing trick: optimized sampling of imbalanced datasets-a brief survey of the recent state of the art. *Eng Rep (Hoboken)*. 2020. <https://doi.org/10.1002/eng2.12298>.
22. Barella VH, García LPF, Souto MD, Lorena AC, Carvalho A. Assessing the data complexity of imbalanced datasets. *Inf Sci*. 2021;553:83–109. <https://doi.org/10.1016/j.ins.2020.12.006>.
23. Wu F, Jing X, Shan S, Zuo W, Yang J-y. Multiset feature learning for highly imbalanced data classification. *IEEE Trans Pattern Anal Mach Intell*. 2021;43:139–56.
24. Tao X, Li Q, Ren C, Guo W, Li C, He Q, et al. Real-value negative selection over-sampling for imbalanced data set learning. *Expert Syst Appl*. 2019;129:118–34. <https://doi.org/10.1016/J.ESWA.2019.04.011>.
25. Mathews L, Hari S. Learning from imbalanced data. *Advances in Computer and Electrical Engineering*. 2019. <https://doi.org/10.4018/978-1-5225-2255-3.CH159>.
26. Zhao J, Jin J, Chen S, Zhang R, Yu B, Liu Q. A weighted hybrid ensemble method for classifying imbalanced data. *Knowl-Based Syst*. 2020;203:106087. <https://doi.org/10.1016/j.knosys.2020.106087>.
27. Bekkar M, Djemaa H, Alitouche TA. Evaluation measures for models assessment over imbalanced data sets. *Journal of Information Engineering and Applications*. 2013;3:27–38.
28. Basha SJ, Madala S, Vivek K, Kumar ES, Ammannamma T. A review on imbalanced data classification techniques. 2022 International Conference on Advanced Computing Technologies and Applications (ICACTA), 1–6 (2022) <https://doi.org/10.1109/ICACTA54488.2022.9753392>
29. Öztürk MM. Which type of metrics are useful to deal with class imbalance in software defect prediction? *Inf Softw Technol*. 2017;92:17–29.
30. Cruz Huayanay A, Bazán JL, Russo CM. Performance of evaluation metrics for classification in imbalanced data. *Comput Stat*. 2025;40(3):1447–73.
31. Fawcett T. An introduction to roc analysis. *Pattern Recognit Lett*. 2006;27(8):861–74.
32. García V, Mollineda RA, Sánchez JS. A bias correction function for classification performance assessment in two-class imbalanced problems. *Knowledge-Based Systems*. 2014;59:66–74.
33. Jiménez-Navarro M, Troncoso-García A, Troncoso A, Martínez-Álvarez F, Martínez-Ballesteros M. Explainable deep learning with embedded feature selection for electricity demand forecasting. In: *2024 International Conference on Smart Systems and Technologies (SST)*, pp. 153–158 (2024). IEEE.
34. Troncoso-García AR, Martínez-Ballesteros M, Martínez-Álvarez F, Troncoso A. A new metric based on association rules to assess feature-attribution explainability techniques for time series forecasting. *IEEE Trans Pattern Anal Mach Intell*. 2025. <https://doi.org/10.1109/TPAMI.2025.3540513>.
35. Troncoso-García A, Martínez-Ballesteros M, Martínez-Álvarez F, Troncoso A. Explainable machine learning for sleep apnea prediction. *Procedia Comput Sci*. 2022;207:2930–9.
36. Kadir MA, Mosavi A, Sonntag D. Evaluation metrics for xai: A review, taxonomy, and practical applications. In: *2023 IEEE 27th International Conference on Intelligent Engineering Systems (INES)*, pp. 000111–000124 (2023). IEEE.
37. Wong T-T, Chung P-C. A consistency analysis on four evaluation metrics for classifying imbalanced data. *Knowledge and Information Systems*, 2025; 1–18.
38. Mahmud Sujon K, Binti Hassan R, Tusnia Towshi Z, Othman MA, Abdus Samad M, Choi K. When to use standardization and normalization: empirical evidence from machine learning models and xai. *IEEE Access*. 2024;12:135300–14. <https://doi.org/10.1109/ACCESS.2024.3462434>.
39. Bolton RJ, Hand DJ. Statistical fraud detection: a review. *Stat Sci*. 2002;17(3):235–55.
40. Khalilia M, Chakraborty S, Popescu M. Predicting disease risks from highly imbalanced data using random forest. *BMC Med Inform Decis Mak*. 2011;11:1–13.

41. Gilmore E, Estivill-Castro V, Hexel R. More interpretable decision trees. In: Hybrid Artificial Intelligent Systems: 16th International Conference, HAIS 2021, Bilbao, Spain, September 22–24, 2021, Proceedings 16, pp. 280–292 (2021). Springer
42. Chen T, Guestrin C. Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining, 2016;pp. 785–794
43. Halder RK, Uddin MN, Uddin MA, Aryal S, Khraisat A. Enhancing k-nearest neighbor algorithm: a comprehensive review and performance analysis of modifications. *J Big Data*. 2024;11(1):113.
44. Brownlee J. Failure of classification accuracy for imbalanced class distributions. *Machine Learning Mastery*. 2020;31.
45. Najem SM, Kadeem SM. A survey on fraud detection techniques in e-commerce. *Tech-Knowledge*. 2021;1(1):33–47.
46. Zeynali Tazehkandi M, Nowkarizi M. Three approaches to measuring recall on the web: a systematic review. *Electron Libr*. 2020;38(3):477–92.
47. Jeni LA, Cohn JF, De La Torre F. Facing imbalanced data—recommendations for the use of performance metrics. In: 2013 Humaine Association Conference on Affective Computing and Intelligent Interaction, pp. 245–251 (2013). IEEE
48. Chicco D, Jurman G. The advantages of the matthews correlation coefficient (MCC) over f1 score and accuracy in binary classification evaluation. *BMC Genomics*. 2020;21:1–13.
49. Alsulmi M. From ranking search results to managing investment portfolios: exploring rank-based approaches for portfolio stock selection. *Electronics*. 2022;11(23):4019.
50. Alsubaie Y, El Hindi K, Alsalmán H. Cost-sensitive prediction of stock price direction: selection of technical indicators. *IEEE Access*. 2019;7:146876–92.
51. Szegehalmy S, Fazekas A. A comparative study of the use of stratified cross-validation and distribution-balanced stratified cross-validation in imbalanced learning. *Sensors*. 2023;23(4):2333.
52. Pembury Smith MQ, Ruxton GD. Effective use of the mcnemar test. *Behav Ecol Sociobiol*. 2020;74:1–9.
53. Aguirre-Urreta MI, Rönkkö M. Statistical inference with plsc using bootstrap confidence intervals. *MIS Q*. 2018;42(3):1001–10.
54. Więckowska B, Kubiak KB, Józwiak P, Moryson W, Stawińska-Witoszyńska B. Cohen's kappa coefficient as a measure to assess classification improvement following the addition of a new marker to a regression model. *Int J Environ Res Public Health*. 2022;19(16):10213.
55. Chase Lipton Z, Elkan C, Narayanaswamy B. Thresholding classifiers to maximize f1 score. *arXiv e-prints*, 2014; 1402
56. Jiang J, Jiang X, Xu L, Zhang Y, Zheng Y, Kong D. Noise-robustness test for ultrasound breast nodule neural network models as medical devices. *Front Oncol*. 2023;13:1177225.
57. Knief U, Forstmeier W. Violating the normality assumption may be the lesser of two evils. *Behav Res Methods*. 2021;53(6):2576–90.
58. Yeh I-C, Lien C-h. Default of Credit Card Clients Dataset. UCI Machine Learning Repository 2009; <https://doi.org/10.24432/C55S3H>.
59. Azis H. Assessing the performance of logistic regression in heart disease detection through 5-fold cross-validation. *International Journal of Artificial Intelligence in Medical Issues*. 2024;2(1):1–11.
60. Maina DG, Moso JC, Gikunda PK. Detecting fraud in motor insurance claims using xgboost algorithm with smote. In: 2023 International Conference on Information and Communication Technology for Development for Africa (ICT4DA), 2023;pp. 61–66. IEEE
61. Basak S, Huber M. Evolutionary feature scaling in k-nearest neighbors based on label dispersion minimization. In: 2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2020; pp. 928–935. IEEE
62. Kusa W, Peikos G, Staudinger M, Lipani A, Hanbury A. Normalised precision at fixed recall for evaluating tar. In: Proceedings of the 2024 ACM SIGIR International Conference on Theory of Information Retrieval, 2024; pp. 43–49
63. Foody GM. Challenges in the real world use of classification accuracy metrics: from recall and precision to the matthews correlation coefficient. *PLoS ONE*. 2023;18(10):0291908.
64. Tang J, Li Y, Hou Z, Fu S, Tian Y. Robust two-stage instance-level cost-sensitive learning method for class imbalance problem. *Knowledge-Based Systems*. 2024;300:112143.
65. Chen X, Chen P. A comparison of four methods for the analysis of n-of-1 trials. *PLoS ONE*. 2014. <https://doi.org/10.1371/journal.pone.0087752>.
66. Owusu-Adjei M, Hayfron-Acquah JB, Frimpong T, Abdul-Salaam G. Imbalanced class distribution and performance evaluation metrics: a systematic review of prediction accuracy for determining model performance in healthcare systems. *PLoS Digit Health*. 2023. <https://doi.org/10.1371/journal.pdig.0000290>.
67. Wardhani NWS, Rochayani MY, Iriany A, Sulistyono A, Lestantyo P. Cross-validation metrics for evaluating classification performance on imbalanced data. 2019 International Conference on Computer, Control, Informatics and its Applications (IC3INA), 14–18 (2019) <https://doi.org/10.1109/IC3INA48034.2019.8949568>
68. Explaining xgboost predictions with shap value. A comprehensive guide to interpreting decision tree-based models. *New Trends in Computer Sciences*. 2023. <https://doi.org/10.3846/ntcs.2023.17901>.
69. Mokhtari KE, Higdon BP, Başar A. Interpreting financial time series with shap values. In: Proceedings of the 29th Annual International Conference on Computer Science and Software Engineering, 2019;pp. 166–172

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